ViRefSAM: Visual Reference-Guided Segment Anything Model for Remote Sensing Segmentation

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***Abstract*—The segment anything model (SAM), with its prompt-driven paradigm, exhibits strong generalization in generic segmentation tasks. However, applying SAM to remote sensing (RS) images still faces two major challenges. First, manually constructing precise prompts for each image (e.g., points or boxes) is labor-intensive and inefficient, especially in RS scenarios with dense small objects or spatially fragmented distributions. Second, SAM lacks domain adaptability, as it is pretrained primarily on natural images and struggles to capture RS-specific semantics and spatial characteristics, especially when segmenting novel or unseen classes. To address these issues, inspired by few-shot learning (FSL), we propose ViRefSAM, a novel framework that guides SAM utilizing only a few annotated reference images that contain class-specific objects. Without requiring manual prompts, ViRefSAM enables automatic segmentation of class-consistent objects across RS images. Specifically, ViRefSAM introduces two key components while keeping SAM's original architecture intact: 1) a visual contextual prompt (VCP) encoder that extracts class-specific semantic clues from reference images and generates object-aware prompts via contextual interaction with target images and 2) a dynamic target alignment (DTA) adapter, integrated into SAM's image encoder, which mitigates the domain gap by injecting class-specific semantics into target image features, enabling SAM to dynamically focus on task-relevant regions. Extensive experiments on three few-shot segmentation (FSS) benchmarks, including iSAID-5^i, LoveDA-2^i, and COCO-20^i, demonstrate that ViRefSAM enables accurate and automatic segmentation of unseen classes by leveraging only a few reference images, and consistently outperforms existing FSS methods across diverse datasets.**

***Index Terms*—Few-shot learning (FSL), remote sensing (RS), segment anything model (SAM), semantic segmentation.**

I. Introduction

SEMANTIC segmentation is a fundamental task in remote sensing (RS) intelligent interpretation systems [1], [2], [3], aimed at assigning predefined semantic labels to each pixel in RS images for accurate identification and spatial localization of land cover categories. This task plays a critical role in various real-world applications, including land use monitoring [4], [5], urban planning [6], [7], and disaster assessment [8], [9]. With the rapid development of deep learning, particularly with the emergence of fully convolutional networks (FCNs) [10], [11], [12] and Transformer-based architectures [13], [14], [15], [16], the accuracy and generalization capabilities of RS semantic segmentation have substantially improved, providing a solid foundation for the automated analysis of RS images.

The segment anything model (SAM) has recently emerged as a highly influential vision foundation model for generic segmentation [17], [18], [19], [20]. Pretrained on billions of labeled samples, SAM demonstrates impressive zero-shot generalization across diverse segmentation tasks. Its core innovation is the prompt-driven paradigm, which allows users to specify target regions flexibly using point, bounding box, or mask prompts for efficient and accurate segmentation [see Fig. 1(a)]. Motivated by SAM's capabilities, researchers have begun applying it to RS semantic segmentation [21], [22], [23], [24], [25]. For example, Wang et al. [21] proposed an efficient pipeline that combines SAM with existing RS object detection datasets to generate large-scale RS segmentation datasets. Similarly, RingMo-SAM [23] proposed a multimodal RS segmentation foundation model that segments objects in optical and SAR data while also identifying object categories. SSRS [24] introduced object consistency and boundary preservation losses to refine SAM's output for segmentation tasks.

Despite its promising performance, the application of SAM to RS imagery still faces several key challenges.

1) *Prompt Construction Efficiency:* SAM requires manually constructing precise and reasonable prompts (e.g.,

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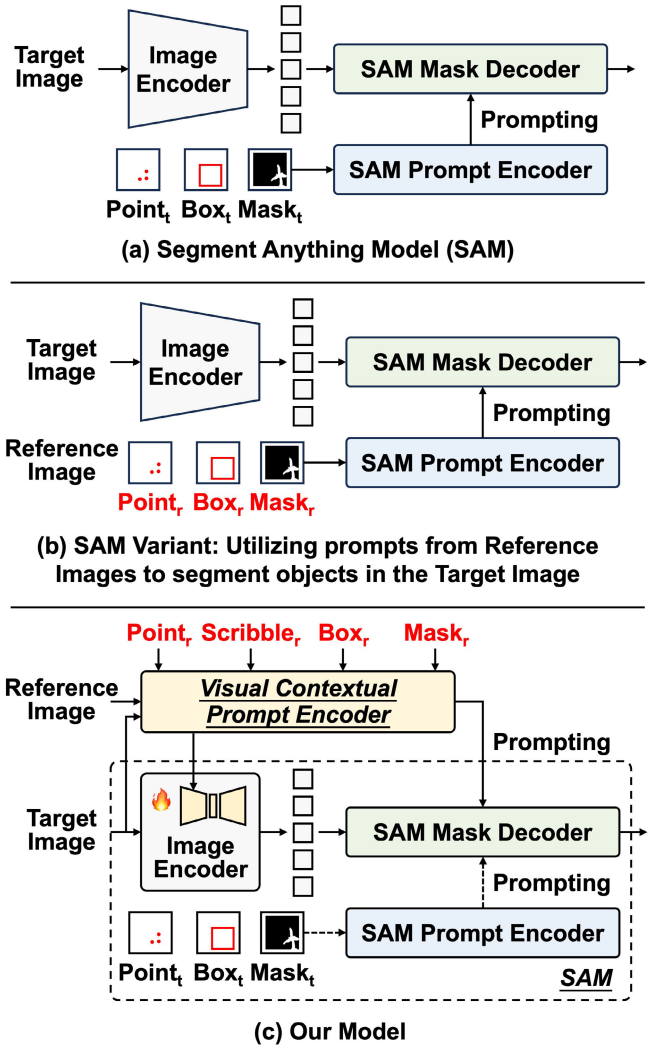


Fig. 1. Framework comparison among SAM, SAM variant, and our ViRefSAM. (a) Original SAM: requires manually customized prompts (e.g., points, boxes, or masks) for each individual image, which must be carefully constructed to activate the correct target. (b) *SAM variant:* attempts to reduce interaction cost by reusing prompts (e.g., a box or point) defined in one reference image across other images. However, this scheme lacks semantic alignment between images and may fail under significant variations in object appearance and background. (c) *Our ViRefSAM:* introduces a VCP Encoder to automatically generate object-aware prompts by encoding high-level semantic information from the reference image and dynamically interacting with the target image features. This design eliminates the need for manually customized or reused prompts while preserving SAM's core architecture.

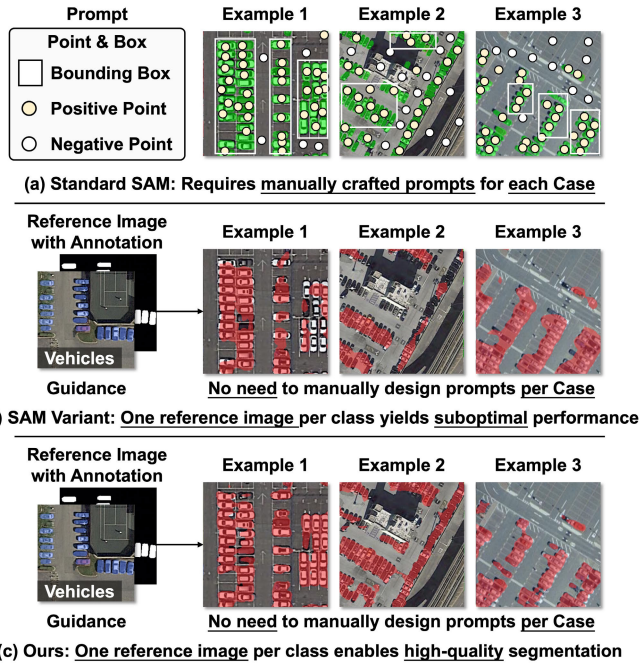


Fig. 2. Comparison of segmentation workflow among SAM, SAM variant, and our ViRefSAM. (a) *SAM:* requires manually crafted prompts (e.g., points or boxes) for each image, making it inefficient in large-scale RS segmentation. (b) *SAM variant:* reuses prompts generated from a single labeled reference image per class, but lacks task-specific alignment, resulting in limited segmentation quality. (c) *Our ViRefSAM:* introduces a VCP encoder and DTA adapter to fully exploit annotated reference images, enabling the accurate and automatic segmentation of specific classes across RS images.

points, boxes, or rough masks) for each RS image, which is both time-consuming and labor-intensive [see Fig. 2(a)]. To segment specific objects, users must first accurately localize targets within complex scenes and then provide the corresponding prompts to activate SAM. The more precise and reasonable these prompts are, the more effective the segmentation. This process becomes particularly cumbersome in RS scenarios involving challenging spatial structures and dense small objects, such as "vehicles" densely packed in parking lots, "ships" scattered across open waters, or "buildings" in urban residential areas. These targets often exhibit high intraclass variability, tiny object scale, and cluttered or fragmented spatial distributions, significantly increasing the manual annotation burden.

2) *Domain-Specific Knowledge:* Since SAM is predominantly pretrained on natural scene images, it lacks the domain-specific semantic and spatial understanding required for RS imagery, leading to degraded segmentation performance. This limitation becomes more pronounced when encountering novel or unseen RS-specific classes not present in the pretraining corpus, restricting its generalization in real-world applications.

To enhance both efficiency and generalization of SAM in RS segmentation, the few-shot learning (FSL) paradigm [26], [27], [28], [29] offers a promising solution. This paradigm enables models to learn novel patterns from only a few labeled examples. In this context, visual contextual samples (i.e., reference images) can serve as semantic prompts to guide SAM in segmenting desired objects. As illustrated in Fig. 2(b) and (c), by providing a single reference image containing instances of "vehicles," SAM can automatically segment "vehicles" in various RS images without requiring manually crafted prompts for each image. A straightforward engineering implementation is to reuse prompts across images. As illustrated in Fig. 1(b), point or box annotations on a reference image can be encoded by SAM's prompt encoder and directly applied to segment "vehicles" in other images via SAM's mask decoder. However, such static prompt reuse often overlooks the semantic gap

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between RS images caused by significant differences in object appearance, scale, and background complexity, which ultimately hinders generalization performance.

To address these limitations, we propose ViRefSAM, a visual reference-guided segmentation framework that leverages visual reference images of a specific category to automatically segment class-consistent objects across diverse RS images. Unlike previous schemes that rely on manually customized or statically reused prompts, ViRefSAM introduces a visual contextual prompt (VCP) encoder that learns high-level, class-specific semantic representations from the reference image and dynamically interacts with the target image features. This enables the generation of object-aware prompts in a context-sensitive manner, without modifying SAM's core architecture. As illustrated in Fig. 1(c), the reference image and its corresponding annotations (supporting various formats such as points, scribbles, bounding boxes, and dense masks) are fed into the VCP Encoder to generate semantic prompts, which are then forwarded to SAM's mask decoder to activate the desired target objects.

The VCP encoder operates in two primary stages: visual contextual interaction and object-aware prompt generation.

1) *Visual Contextual Interaction:* Initially, the reference image is condensed into a set of reference prototypes guided by the provided annotations, which effectively capture semantic clues of class-specific objects. These prototypes then engage with the target image features (the image to be segmented) via a cross-attention structure, enabling the extraction of class-specific semantic information directly from the target image. The target image's pseudo-mask is subsequently derived by aligning the reference prototypes with the target image features, ensuring accurate object localization.
2) *Object-Aware Prompt Generation:* A set of learnable queries is introduced to perform mask attention, constrained by the generated pseudo-mask, enabling the extraction of object-specific semantics from the target image. These representations culminate in the generation of object-aware embeddings that facilitate segmentation. To enhance the generation quality and prevent semantic redundancy, we incorporate a regularization loss, ensuring that the embeddings are distinct and contextually relevant.

To further enhance domain adaptation for RS target segmentation, we fine-tune SAM within a metalearning framework [30]. The training set is organized into multiple tasks (i.e., episodes), each focusing on a specific object category. This forces the model to learn class-specific contextual information from reference images to segment corresponding objects in target images. For this purpose, we propose a dynamic target alignment (DTA) adapter. Unlike traditional adapters [31] that adjust image features through downsampling and upsampling projections, the DTA adapter dynamically injects class-specific semantic clues from the reference features into the target image representations according to the task category. This injection encourages SAM to emphasize class-relevant regions and suppress background noise. With this fine-tuning strategy, SAM can quickly and accurately localize novel category objects in RS scenes by leveraging reference images, without compromising its generalization ability. Together, the VCP encoder and DTA adapter enable ViRefSAM to effectively embed RS knowledge into both the image encoding and mask decoding stages, resulting in robust segmentation of RS-specific objects.

For evaluation, we followed the standard few-shot segmentation (FSS) protocol [32], [33], [34], which utilizes several reference images to guide the model in segmenting unseen classes. We conducted experiments on two RS FSS datasets, iSAID-5^i and LoveDA-2^i, as well as the CV dataset COCO-20^i. Results show that ViRefSAM effectively mitigates the limitations of manual prompt engineering in SAM, achieving automatic and accurate segmentation of class-specific objects, even for unseen classes, guided by only a few reference images. It outperforms existing FSS methods, including vision foundation models such as PerSAM and Matcher, across all three datasets, delivering the best results overall.

The main contributions can be summarized as follows.

1) We enhance the functionality of SAM by proposing the ViRefSAM framework, which leverages visual reference images of a specific category to automatically segment the class-consistent objects across diverse images, eliminating the need for manually customized prompts for each image.
2) We introduce a VCP encoder that extracts class-specific semantic representations from reference images and dynamically interacts with target image features to generate object-aware prompts, which are seamlessly integrated into SAM's mask decoder for segmentation.
3) We propose a dynamic task alignment adapter, which, in conjunction with the metalearning training paradigm, drives the SAM model to dynamically focus on class-specific objects within the current task, enhancing the generalization capability in RS scenarios.
4) Extensive experiments on RS and CV FSS datasets demonstrate the effectiveness of ViRefSAM. With only a few reference images, it enables automatic segmentation of class-specific objects (even for unseen classes) and outperforms existing FSS methods across multiple datasets.

II. Related Works

A. RS Semantic Segmentation

Semantic segmentation, which assigns a label to each pixel, is essential in RS image interpretation. Early work by Long et al. [10] introduced FCNs, laying the foundation for subsequent developments. To capture contextual information, Chen et al. [35] added dilated convolutions, while methods like the pyramid pooling module (PPM) [36] and atrous spatial pyramid pooling (ASPP) [37] enabled multiscale fusion. In RS, challenges such as complex scenes, small targets, and class imbalance are prevalent. To address these, Peng et al. [38] proposed CF-Net for small-object representation, Feng et al. [39] introduced NPALoss for boundary refinement, and Niu et al. [40] used a disentangled framework

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for foreground and boundary modeling. While CNN-based methods perform well on large datasets, their generalization to unseen categories and domains, especially in low-data conditions, remains limited.

Recently, Transformer-based models, like SegFormer [41] and Mask2Former [15], have shown strong performance by modeling long-range dependencies. RS studies have adapted these models with specialized architectures [14], [42], [43], [44], such as UNetFormer [14] for urban segmentation and hybrid CNN–Transformer models for multiscale feature learning. However, these approaches still require extensive annotations, limiting their use in few-shot and cross-domain tasks.

B. Segment Anything Model

The SAM [17], developed by Meta AI, is a universal interactive segmentation framework built on the vision transformer architecture. Pretrained on billions of labeled samples, SAM demonstrates strong zero-shot generalization across diverse segmentation tasks. It consists of an image encoder for extracting visual features, a prompt encoder that converts user inputs (e.g., points, boxes, or masks) into embeddings, and a mask decoder that generates segmentation masks. SAM's versatility has led to its adoption in fields beyond traditional semantic segmentation, such as MedSAM [45] for medical imaging, inpaint anything [46] and edit everything [47] for image editing, and object tracking [48], [49], where it improves localization robustness with prompt-guided initialization. These applications underscore SAM's potential as a vision foundation model, particularly in data-scarce or weakly supervised settings.

C. SAM in RS

In the RS domain, SAM has gained increasing attention for reducing annotation costs and enhancing segmentation quality in large-scale Earth observation. Existing studies mainly adapt SAM with RS-specific priors or leverage its features for downstream tasks. MeSAM [59] introduces multiscale enhancement to address boundary and texture limitations, while RingMo-SAM [23] and RSPrompter [22] integrate multimodal and category-aware priors for improved segmentation. More recent works, such as ClassWise-SAM-Adapter [60], ROS-SAM [61], and AerOSeg [62], enhance SAM through adapter-based fine-tuning, LoRA optimization, or open-vocabulary alignment. Similarly, SESSRS [25] and MSA-SAM [63] demonstrate that multiscale adaptation and SAM-based postprocessing can refine segmentation performance without retraining the backbone. Beyond visible-spectrum segmentation, several efforts extend SAM to small-object and multimodal perception. IRSAM [64] redesigns SAM's encoder–decoder with Perona–Malik diffusion and a granularity-aware decoder for infrared small-target detection. IOVarNet [65] employs an inner–outer variational synergy network with total variation modules and dual-space attention to enhance spatial responses and target discrimination. STAMF [66] fuses RGB and polarization cues via cross-modal attention for robust underwater small-object detection. These studies highlight the growing trend of adapting SAM for fine-grained and multimodal understanding.

Despite these advances, most SAM-based RS approaches remain centered on interactive segmentation, open-vocabulary recognition, or structural adaptation, with limited attention to efficiency-oriented segmentation from a few-shot perspective. In contrast, ViRefSAM is designed to overcome SAM's efficiency bottleneck by enabling automatic, class-consistent segmentation across images using only a single annotated reference. By leveraging reference-guided contextual understanding, ViRefSAM substantially reduces dependence on manual prompts while maintaining strong generalization to unseen categories, offering an efficient and scalable framework within the evolving landscape of SAM-based RS research.

D. Few-Shot (Visual Reference) Segmentation

Recent advancements in FSL have aimed at improving generalization to unseen domains, enabling models to effectively classify novel classes with only a few labeled samples. The majority of current methods adopt a metalearning paradigm [30], where transferable knowledge is learned from a series of tasks (episodes) drawn from the training dataset and generalized to novel tasks. FSS extends FSL to dense prediction tasks, aiming to segment unseen classes in target images using a small number of labeled samples (i.e., reference images) [67].

FSS approaches can be grouped into three primary branches:

1) Prototype matching-based methods [see Fig. 3(a)] [32], [33], [34], [50], [51], which generate category-representative prototypes by compressing reference features through mask average pooling [50]. For example, SSPNet [51] aggregates prototypes from the semantic clues within the target image to guide its segmentation. DCPNet [34] refines reference features into multiple prototypes, each representing distinct properties.
2) Feature fusion-based methods [see Fig. 3(b)] [52], [53], [54], [68], which combine features from both the reference and target images. For instance, HSNet [52] utilizes 4-D convolution for multiscale fusion, while IPMT [69] merges reference- and target-image semantics into an intermediate prototype.
3) Pixel matching-based methods [see Fig. 3(c)] [55], [56], [57], [58], which focus on pixel-level similarity between reference and target images. CyCTR [55] uses cross-attention to aggregate pixel-level semantics from the reference image. Recently, VLMs have been explored for FSS tasks [70], [71], [72]. PI-CLIP [70] uses CLIP's semantic alignment to locate target categories, while MVLPNet [72] captures text embeddings for specific categories, aligning them with visual features using an optimal transport scheme to handle intraclass variation in RS imagery.

In the RS domain, FSS has attracted substantial attention [27], [73], [74], [75], [76], [77]. Yao et al. [73] proposed a scale-aware prototype matching approach to tackle the variability in object appearance and scale. Bi et al. [27] developed DMNet to extract class-specific semantics from target images. Peng et al. [78] introduced a multigranularity aggregation

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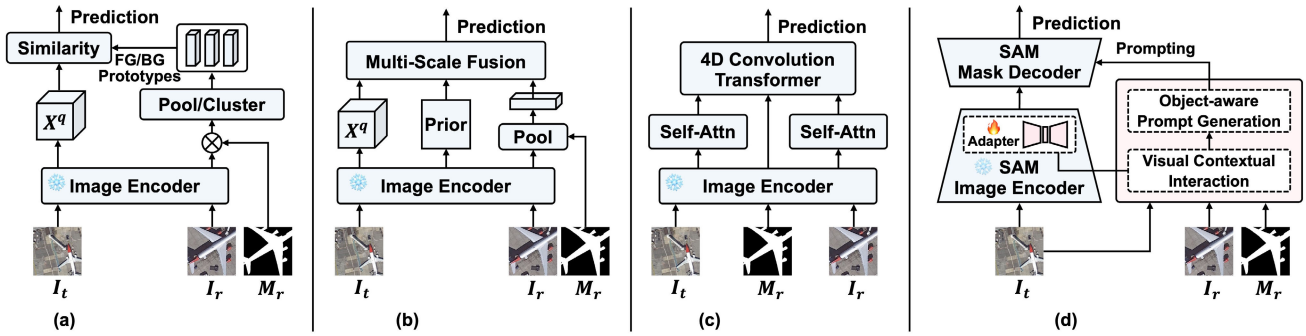


Fig. 3. Overview of typical FSS paradigms. (a) *Prototype matching-based methods* [33], [50], [51] compute class prototypes from reference images and compare them with target features. (b) *Feature fusion-based methods* [52], [53], [54] aggregate reference and target features to enhance class-specific object representation. (c) *Pixel matching-based methods* [55], [56], [57], [58] establish pixelwise correlations between reference and target images for fine-grained segmentation. (d) *Our ViRefSAM*, introduces a VCP encoder into SAM, leveraging reference images to generate object-aware prompts and enabling automatic segmentation of class-specific objects in RS imagery. This method effectively overcomes the manual prompt design burden in SAM-based RS segmentation tasks.

network to address the challenges of intraclass inconsistency in RS FSS tasks. Inspired by these advances, we introduce a VCP encoder into SAM, enabling automatic segmentation of RS objects guided by reference images and mitigating the need for costly manual prompt design.

III. Problem Definition

Existing SAM requires manual prompt customization for each image when segmenting specific objects, which becomes time-consuming and labor-intensive in complex RS scenarios. To address this, inspired by FSS, which utilizes a few labeled samples to guide the model in segmenting novel classes, we enhance SAM by incorporating reference images of specific categories. This enables SAM to automatically segment the same class of objects across multiple target images without needing manual prompts for each image. Therefore, we adopt the task definition from FSS. Following existing FSS methods [27], [34], we employ the metalearning paradigm to train the model. Concretely, given a training dataset D_{train} containing seen classes C_{seen} and a testing dataset D_{test} with unseen classes C_{unseen} (where $C_{\text{seen}} \cap C_{\text{unseen}} = \emptyset$), the model learns transferable metaknowledge from a series of tasks sampled from D_{train} and generalizes directly to D_{test}. Both datasets D_{train} and D_{test} consist of multiple episodes, each containing a reference set $R = \{(X_r^i, M_r^i)\}_{i=1}^K$ and a target set $T = \{(X_t, M_t)\}$. Here, X^i and M^i represent the original image and the binary mask containing the target class, respectively, and K is the number of reference images provided in R. During each training episode, the model is optimized by predicting the target image X_t based on the reference set R. After training, the model is evaluated on the testing episodes in D_{test} without further optimization, with the target mask M_t in T not being provided during the testing phase.

A. Reference Image Annotation Diversity

To ensure annotation diversity, the reference image annotation M_r supports multiple input forms, including points, scribbles, bounding boxes, and masks. Point annotations provide basic localization by placing a single point on the target object. Scribbles allow users to draw freehand curves within the object, offering flexibility in defining boundaries. Bounding boxes enclose the object with a rectangle, enabling quick and rough localization, especially in large or cluttered scenes. Masks provide pixel-level annotations for precise object delineation, ensuring high segmentation accuracy. These diverse annotation types enable our SAM to utilize varying degrees of reference images, balancing efficiency and accuracy according to task requirements.

IV. Proposed Method

A. Method Overview

To address the high cost of manual prompt customization in SAM for RS object segmentation, we propose ViRefSAM, a novel framework that augments SAM with visual reference images containing class-specific objects, without modifying its core architecture. Under the guidance of these reference images, ViRefSAM enables automatic segmentation of target objects in RS images, eliminating the need to manually craft prompts for each image. Specifically, we introduce a VCP encoder that transforms the reference image into object-aware embeddings, which are then injected into SAM's mask decoder to guide the segmentation of the desired object (see Fig. 4). To further enhance SAM's task adaptability in RS scenarios, we incorporate a DTA adapter into the SAM image encoder. This adapter injects class-specific semantic clues from the reference image into the encoded features of the target image, enabling SAM to dynamically focus on task-relevant objects. With this design, ViRefSAM requires only a single reference image from a given category to automatically segment all instances of that category across diverse RS images, significantly improving segmentation efficiency and reducing manual effort.

The subsequent sections detail the core components of ViRefSAM: Section IV-B presents the VCP Encoder, Section IV-C introduces the DTA adapter, and Section IV-D outlines the segmentation prediction process and loss function design for model training.

B. VCP Encoder

As depicted in Fig. 4, we incorporate visual reference images containing specific object categories to guide SAM in

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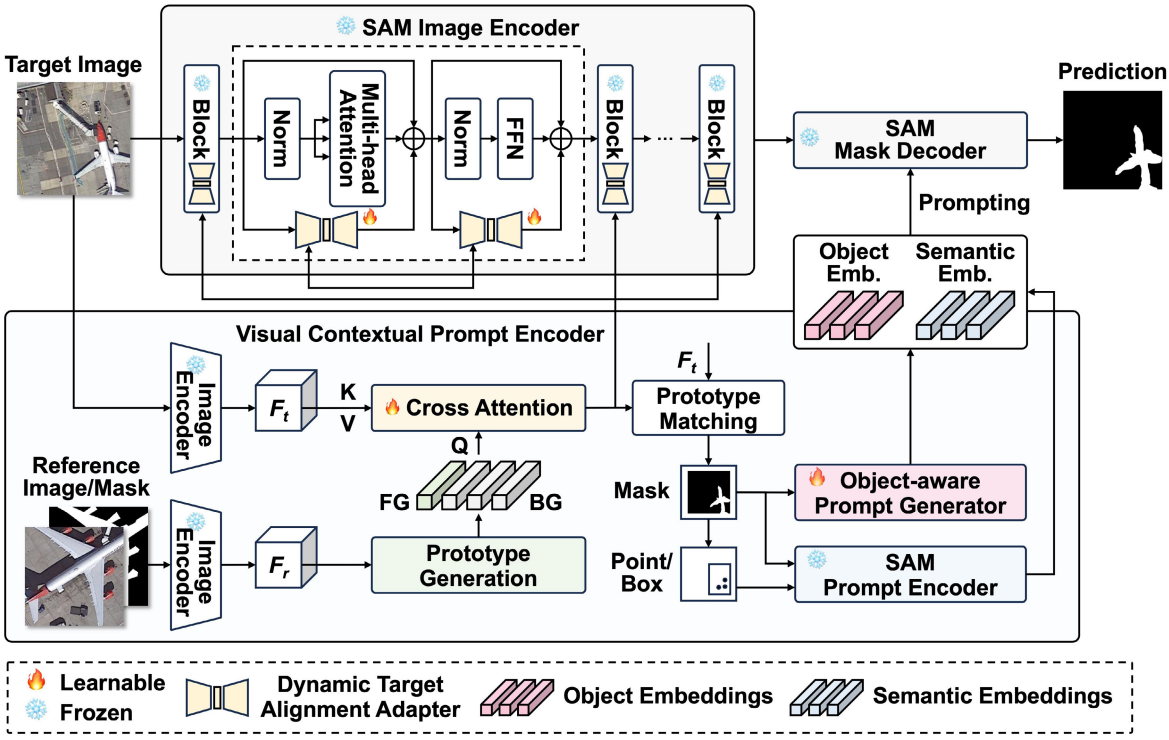


Fig. 4. Overall framework of the proposed method ViRefSAM. Without altering the original architecture of SAM, ViRefSAM introduces the reference image annotated with a specific object category and employs a VCP encoder to extract object-aware embeddings (denoted by "Emb." in this figure), which are injected into SAM's mask decoder to guide segmentation. In parallel, a DTA adapter is integrated into the SAM image encoder to embed class-specific semantics from the reference image into the target image features, enabling SAM to dynamically focus on task-relevant objects. With this design, SAM can automatically segment all instances of the same category across different RS images using only a single reference image, eliminating the need for manual prompt customization.

automatically segmenting all instances of the same category across different RS images. To this end, we design a VCP encoder, which encodes reference images into class-specific embeddings that are subsequently fed into SAM's mask decoder to produce segmentation results. However, directly transferring semantic features from reference images into SAM's mask decoder is often suboptimal, since RS scenes exhibit substantial variations in appearance, illumination, and viewing angle between reference and target images of the same category, leading to a pronounced semantic gap. Previous studies [27], [69] have demonstrated that exploiting the target image's own semantics can effectively enhance segmentation performance.

Following this insight, the VCP Encoder is designed around the principle of mining semantic information from the target image itself and comprises two key stages: visual contextual interaction and object-aware prompt generation. The first stage models cross-image feature interactions between the reference and target images to produce a pseudo-mask that roughly indicates the target object's location. The second stage introduces a set of learnable queries that, within a masked-attention mechanism, use this pseudo-mask as a spatial constraint to extract class-specific semantics from the target image (i.e., within the pseudo-mask regions). These object-aware queries are then passed into SAM's mask decoder as dynamic prompts, enabling more accurate and context-aware segmentation.

1) Visual Contextual Interaction: Given the target image $X_t \in \mathbb{R}^{3\times h\times w}$ and the reference image $X_r \in \mathbb{R}^{3\times h\times w}$, both are fed into the frozen image encoder to generate the target features $F_t \in \mathbb{R}^{C\times H\times W}$ and reference features $F_r \in \mathbb{R}^{C\times H\times W}$, respectively. Next, to derive the class-specific semantics from the reference features, we perform the prototype generation process. Specifically, we separate the foreground and background regions of the reference image utilizing its mask. Given the complex and fragmented nature of backgrounds in RS scenarios, we adopt a Voronoi-based method [79] to divide the background into multiple local regions, resulting in one foreground region $R_f \in \{0,1\}^{H\times W}$ and several background local regions $\{R_b^i\}_{i=1}^{N_b}$. Consequently, through masked average pooling, we can obtain both the foreground prototype $P_f \in \mathbb{R}^{C\times 1}$ and background prototypes $\{P_b^i\}_{i=1}^{N_b} \in \mathbb{R}^{C\times N_b}$ of the reference image

$$P_f = \frac{1}{|R_f|}\sum_{n=1}^{H\times W} F_{r;n}R_{f;n}$$
$$P_b^i = \frac{1}{|R_b^i|}\sum_{n=1}^{H\times W} F_{r;n}R_{b;n}^i, \quad i = 1,2,\ldots,N_b \tag{1}$$

where $F_r;n$ denotes the nth pixel feature of the reference features. R_b^i and P_b^i represent the ith background local region and generated background prototype of the reference image, respectively.

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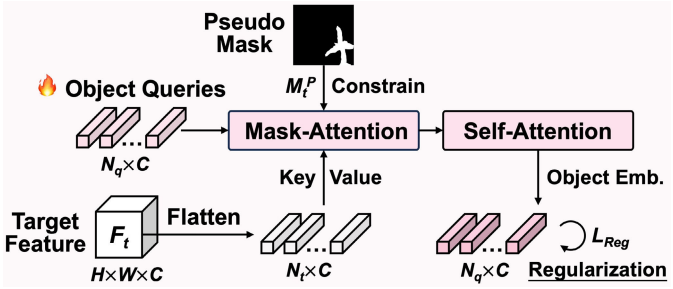


Fig. 5. Detailed pipeline of the object-aware prompt generator. A set of learnable queries is initialized and enhanced by the reference prototype and then refined through mask attention under pseudo-mask constraints to extract object-specific semantics from the target image. A self-attention layer further aligns these embeddings with SAM's feature space. To improve the quality and avoid semantic redundancy, a regularization loss is applied to encourage diversity among the generated embeddings.

To further capture class-specific clues from the target image itself, we utilize the reference prototypes to guide the mining process, as they already contain certain class-specific semantics. Specifically, we employ a cross-attention mechanism to aggregate semantics from the target image, where the reference prototypes $P \in \mathbb{R}^{C\times(N_b+1)}$ serve as the query, and the target features $F_t \in \mathbb{R}^{C\times(HW)}$ act as the key and value

$$\widetilde{P} = \text{Softmax}\left(\frac{(PW_q)(F_tW_k)^T}{\sqrt{d}}\right)(F_tW_v) \tag{2}$$

where $\widetilde{P} = [\widetilde{P}_f, \widetilde{P}_b^1, \ldots, \widetilde{P}_b^{N_b}]$ denote the aligned prototypes, $\widetilde{P}_b^i$ denotes the ith background prototype, F_t denotes the target image features, and W_* denote the projection weights.

After incorporating the class-specific semantics from the target image, we perform prototype matching by computing the similarity between the target image features F_t and the reference prototypes $\widetilde{P}$ (which include a foreground prototpye $\widetilde{P}_f$ and N_b background prototpyes $\{\widetilde{P}_b^i\}_{i=1}^{N_b}$) to obtain its pseudo-mask $M_t^P \in \mathbb{R}^{1\times H\times W}$, which can be formulated as follows:

$$M_t^P = \left[\underset{i}{\text{Max}}\left(\bigcup_{i=1}^{N_b}\langle F_t, \widetilde{P}_b^i\rangle\right); \langle F_t, \widetilde{P}_f\rangle\right] \tag{3}$$

where $\langle\cdot,\cdot\rangle$ denotes the similarity computation and $\bigcup$ denotes the concatenation of multiple segmentation results. Notably, the max operation $\text{Max}(\cdot)$ is utilized to identify the background local prototype most similar to the target image pixels, effectively distinguishing between foreground and background pixels in the target image.

The pseudo-mask M_t^P produced through visual contextual interaction serves as a crucial intermediate representation, providing coarse spatial guidance for locating potential target regions in the target image. This pseudo-mask is then utilized to guide the subsequent object-aware prompt generation, where class-specific semantics are further extracted and transformed into dynamic prompts for SAM's mask decoder.

Object-aware prompt generation aims to produce a set of prompt embeddings for the SAM mask decoder, guided by the visual reference image. As illustrated in Fig. 5, we introduce a set of learnable queries $Q \in \mathbb{R}^{C\times N_q}$, which are initialized utilizing Xavier initialization (with zero mean). These queries are further enhanced by adding the foreground prototype $\widetilde{P}_f$, thereby injecting class-specific semantic clues into the query representation. Subsequently, the queries perform mask attention under the constraint of the pseudo mask M_t^P derived from the visual contextual interaction, enabling them to absorb additional class-specific information from the target image X_t. Finally, a self-attention layer is applied to refine the queries and align them with the SAM representation space, yielding a set of object-aware prompt embeddings suitable for guiding segmentation. The entire process can be formulated as follows:

$$P_o = \text{SelfAttn}\left(\text{MaskAttn}\left(Q, F_t, M_t^P\right)\right) \tag{4}$$

where $P_o \in \mathbb{R}^{C\times N_q}$ denotes the object-aware prompt embeddings.

To improve the quality of object-aware prompt embeddings and avoid semantic redundancy, we introduce a regularization loss L_{reg} that encourages these embeddings to be mutually decorrelated while maintaining consistency in representing the same target object. Specifically, given the prompt embedding set $\{P_o^i\}_{i=1}^{N_q} \in \mathbb{R}^C$, the loss is defined as follows:

$$\mathcal{L}_{\text{reg}} = \frac{\sum_{i=1}^{N_q}\sum_{j\neq i}^{N_q}\left\langle P_o^i, P_o^j\right\rangle}{N_q * (N_q - 1)} \tag{5}$$

where $\langle\cdot,\cdot\rangle$ denotes the inner product between prompt embeddings. This soft regularization encourages the embeddings to be mutually distinctive while remaining semantically aligned with the same object category. By promoting diversity without enforcing strict orthogonality, it mitigates redundancy and enhances the expressiveness of the prompt representation, ultimately leading to more accurate and robust segmentation.

In addition, to further enhance the diversity and expressive power of the prompts, besides the proposed object-aware embeddings, we also leverage SAM's own prompt encoder to extract semantic embeddings from the pseudo-mask and its converted points and bounding boxes. These semantic embeddings P_s are jointly fed into SAM's mask decoder along with the object-aware embeddings to facilitate accurate target segmentation.

C. DTA Adapter

Building upon the design in Section IV-B, ViRefSAM significantly improves SAM's efficiency in RS object segmentation. However, since SAM's pretraining largely relies on natural scene data, it lacks domain-specific knowledge for RS tasks, resulting in lower accuracy for segmenting RS objects. To address this, we propose a DTA adapter, which introduces a lightweight module into SAM's image encoder to dynamically inject class-specific semantic clues from reference images during feature encoding, enabling the model to focus on task-relevant target regions. This design allows SAM to incorporate RS-specific knowledge while maintaining its strong general image representation capability.

Specifically, as illustrated in Fig. 4, we insert a DTA adapter in parallel within each block of the SAM image encoder. During training, the parameters of SAM's original image encoder are frozen, and only the parameters of the adapter are

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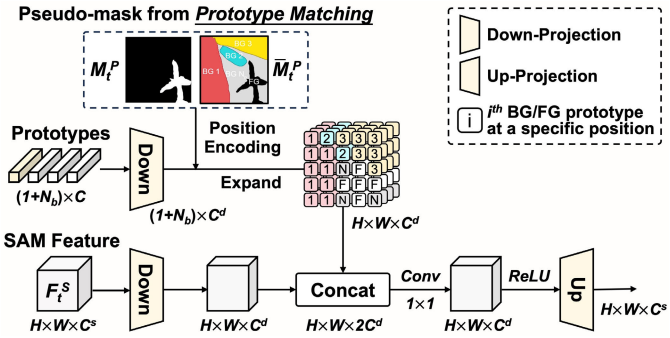


Fig. 6. Detailed pipeline of the DTA adapter. The DTA adapter injects class-specific semantics from the reference image into the encoded features of the target image under positional constraints. This is achieved by embedding the reference prototypes into the target features, enabling SAM to dynamically attend to task-relevant objects during segmentation.

fine-tuned. Given the input target image token x, the output of each block can be expressed as follows:

$$\text{Block}(x) = \text{MultiAttn}(\text{Norm}(x)) + x + \text{DTA}(x) \tag{6}$$

where MultiAttn$(\cdot)$ and Norm$(\cdot)$ are the original structures in the encoder, and DTA$(\cdot)$ is the proposed adapter structure. Next, we provide a detailed introduction to the DTA adapter.

As depicted in Fig. 6, for the features $F_t^S \in \mathbb{R}^{C^s \times H \times W}$ of the target image entering each block, they are first mapped to a C^d-dimensional space via a down-projection layer. Unlike original adapter structures, we extract class-specific clues from the reference image prototypes $\widetilde{P} \in \mathbb{R}^{C \times (1+N_b)}$ and inject them into SAM's features to dynamically focus on the relevant objects for the current task. Specifically, the reference prototypes are also mapped to the C^d-dimensional space utilizing the down-projection layer. The prototypes are then expanded to match the size of the image features for feature concatenation.

To achieve this, we design a dedicated position encoding that ensures the alignment of reference prototypes $\widetilde{P}$ with specific locations in the target image. Since the pseudo-background mask $\bar{M}_t^P$ of the target image, as defined in (3), is derived from the maximum similarity score between multiple reference background prototypes, we can determine the background or foreground prototype ID for each pixel. This guarantees that during the expansion process, different reference prototypes are assigned to the correct pixel positions. This encoding method allows the model to effectively distinguish between foreground and background regions, enhancing segmentation accuracy and ensuring that the appropriate prototype influences the corresponding regions in the image.

After feature concatenation, we utilize a 1×1 convolution to perform feature fusion and apply an up-projection layer to restore the feature dimensions to the C^s-dimensional space, generating the final output of the DTA adapter. The entire process can be formalized as follows:

$$\text{DTA}\left(F_t^S\right) = \left(\text{Conv}\left[F_t^S W_{\text{down}}; \text{Pos}\left(\widetilde{P} W_{\text{down}}\right)\right]\right) W_{\text{up}} \tag{7}$$

where $W_{\text{down}}(\cdot)$ and $W_{\text{up}}(\cdot)$ represent the down- and up-projection layers, respectively. Conv$(\cdot)$ refers to the 1×1 convolution operation utilized for feature fusion, and Pos$(\cdot)$ denotes the position encoding process, which aligns the reference prototypes with specific locations in the target image

$$\text{Pos}\left(\widetilde{P} W_{\text{down}}\right)(i,j) = \begin{cases} \widetilde{P}^m W_{\text{down}}, & \text{if } \bar{M}_t^P(i,j) = m \\ 0, & \text{else.} \end{cases} \tag{8}$$

Here, we present the specific formula for the position encoding process that we have developed, where m denotes the mth reference prototype from $\widetilde{P} \in \mathbb{R}^{C \times (1+N_b)}$.

D. ViRefSAM Prediction

1) One-Shot Setting: Overall, given a target image X_t and a reference image X_r with its associated annotation M_r, the segmentation result $\mathcal{M}_t$ of ViRefSAM can be formulated as follows:

$$\begin{aligned} \mathcal{M}_t &= \text{ViRefSAM}(X_t, X_r, M_r) \\ &= \text{SAMDec}\left(\text{SAMEnc}(X_t), \text{VCPEnc}(X_t, X_r, M_r)\right) \end{aligned} \tag{9}$$

where SAMEnc$(\cdot)$ represents the SAM image encoder equipped with the proposed DTA adapter, VCPEnc$(\cdot)$ denotes the VCP Encoder that generates object-aware embeddings based on both the target and reference inputs, and SAMDec$(\cdot)$ refers to the original SAM mask decoder. This formulation enables the model to generate accurate segmentation masks $\mathcal{M}_t$ under visual contextual guidance without manual prompt design.

2) K-Shot Setting: When extending to the K-shot setting (i.e., $K > 1$), multiple reference images are provided to guide the segmentation. Our ViRefSAM can effectively handle this case by feeding each reference image and its corresponding mask into the VCP Encoder to generate a pair of object-aware prompt embeddings and semantic embeddings. This results in K pairs of embeddings, which are collectively input into SAM's mask decoder to produce the segmentation output. This flexible design benefits from the inherent self-attention mechanism of the SAM architecture.

3) Loss Function: To supervise the training of ViRefSAM, we adopt a combination of binary cross-entropy (BCE) loss and Dice loss. The BCE loss ensures pixel-wise accuracy, while the Dice loss complements it by capturing region-level consistency. The overall loss function of ViRefSAM is defined as follows:

$$\mathcal{L} = \text{BCE}(\mathcal{M}_t, M_t) + \text{Dice}(\mathcal{M}_t, M_t) + \gamma \mathcal{L}_{\text{reg}} \tag{10}$$

where γ denotes the contribution of $\mathcal{L}_{\text{reg}}$.

V. Experiments

To begin with, Section V-A introduces the experimental settings, including dataset descriptions, implementation details, and evaluation metrics. To validate the effectiveness of the proposed method, extensive comparisons with state-of-the-art methods are presented in Section V-B. Finally, Section V-C provides a detailed ablation study to further analyze the contributions of each component in ViRefSAM.

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A. Experimental Setup

1) Datasets: Following the few-shot setting, we conduct comprehensive performance evaluations on three FSS benchmark datasets, including two RS datasets (iSAID-5^i [27] and LoveDA-2^i [27]) and one CV dataset (COCO-20^i [80]).

iSAID-5^i is derived from the iSAID segmentation dataset [88], which contains 655 451 object instances across 15 categories in 2806 high-resolution RS images. Following the DMNet setup [27], the 15 categories are divided into three folds, each with ten training classes and five testing classes. The training and testing classes are mutually exclusive within each fold to simulate the unseen-class scenario. For evaluation, 1000 reference-target pairs are randomly sampled from each fold.

LoveDA-2^i, is built upon the LoveDA dataset [89], consisting of 5 987 high-resolution images and 166 768 labeled objects across urban and rural regions. The dataset includes seven semantic categories (e.g., "road", "water", "barren", and "forest"). Following [27], we exclude the background and split the remaining six classes into three folds, each with four classes for training and two for testing. All images are randomly cropped to 473×473. During evaluation, 1000 reference-target pairs are sampled from each fold. Unlike iSAID, which focuses on instance-level segmentation with specific object categories, LoveDA requires panoramic segmentation of broader land cover classes, making it more challenging due to the larger and continuous nature of the target regions.

COCO-20^i, is constructed from the MSCOCO dataset [90], which contains 80 object classes. Following the protocol in [34], the classes are divided into four folds, each with 60 classes for training and 20 for testing. For each fold, 1000 reference-target pairs are randomly sampled for validation.

2) Implementation Details: The proposed ViRefSAM framework is implemented utilizing the PyTorch library [92] and trained on an NVIDIA Tesla A100 GPU (40G). The model is optimized with the AdamW optimizer, using a batch size of 8 and an initial learning rate of $2e^{-4}$. A cosine learning rate decay schedule is applied throughout training. For the iSAID dataset, we train the model for 70 epochs; for LoveDA-2^i and COCO-20^i, the total number of training epochs is set to 50. All input images are resized to 512×512 to comply with SAM's input requirements. For reference image selection, we follow the standard FSS protocol, where one (or k) reference image is randomly sampled from the same object category as the target image. A fixed random seed is employed to ensure consistency and reproducibility across all experiments. We also adopt the same data augmentation strategies as previous FSS studies.

For the SAM model, we adopt the SAM-Huge architecture. As for the VCP encoder, we experiment with four backbone architectures, including ResNet-50, ResNet-101, ViT-B/16 [13], and DeiT-B/16 [93], all of which are pretrained on ImageNet [94]. This allows direct comparison with various FSS methods utilizing different encoder types. In our experiments, the number of background prototypes N_b is set to 5, the number of learnable queries N_q is set to 64, and the weight γ for the query regularization loss is set to 0.2.

3) Evaluation Metrics: Following prior work in FSS [81], we adopt two commonly used evaluation metrics: mean intersection-over-union (mIoU) and foreground–background IoU (FB-IoU). Specifically, mIoU measures the average segmentation performance across all unseen classes and is defined as: $\text{mIoU} = (1/n)\sum_{i=1}^{N_{\text{unseen}}} \text{IoU}_i$, where N_{unseen} is the number of unseen classes in each evaluation fold and IoUi denotes the IoU score for class i. FB-IoU provides a coarser-grained evaluation by averaging the segmentation performance over foreground and background regions, calculated as: $\text{FB-IoU} = (\text{IoU}_F + \text{IoU}_B)/2$, where IoU_F and IoU_B represent the IoU scores for the foreground and background, respectively. Considering its fine-grained class-level granularity and broader applicability, mIoU is selected as the primary evaluation metric throughout our experiments.

B. Comparison With State of the Arts

1) Quantitative Results: In this section, we present quantitative evaluations on three FSS datasets. Specifically, we compare the proposed ViRefSAM with existing FSS and RS-FSS methods on two RS benchmarks, iSAID-5^i and LoveDA-2^i. Furthermore, we assess the generalization capability of ViRefSAM beyond the RS domain by evaluating its performance alongside several foundation models on the COCO-20^i dataset.

iSAID-5^i: As depicted in Table I, we report the one-shot and five-shot segmentation results of the proposed ViRefSAM on the iSAID-5^i dataset, compared with several competitive methods. It is evident that ViRefSAM consistently achieves state-of-the-art performance under all backbone settings. For example, under the one-shot setting with the ResNet-50 backbone, ViRefSAM attains 53.06% mIoU, outperforming the previous best method MGANet [78] (51.85%). When adopting the DeiT-B/16 backbone, it further achieves 53.13% mIoU, surpassing AgMTR [86] by 1.55%. Similar performance gains are observed across other backbones. Under the five-shot setting, ViRefSAM also demonstrates superior results in both mIoU and FB-IoU metrics. Notably, in some cases, our method under the one-shot setting even outperforms other methods under the five-shot setting, such as R2Net [76] and DMNet [27]. These results highlight the strong segmentation capability and generalization of ViRefSAM, which can accurately identify class-specific objects in RS images under the guidance of reference images, even for unseen classes.

To provide a more detailed analysis, Table II reports the classwise IoU scores of ViRefSAM across all three folds of the iSAID-5^i dataset. Specifically, C1–C5 correspond to the testing classes in Fold 0, C6–C10 to Fold 1, and C11–C15 to Fold 2. The results show that ViRefSAM ranks among the top-performing methods in most categories, achieving competitive or superior performance across diverse object types. Although performance variations can be observed for certain classes, ViRefSAM maintains consistently strong performance across different folds, indicating robust generalization behavior. Overall, these class-wise results suggest that ViRefSAM exhibits strong generalization ability in fine-grained RS segmentation tasks under limited supervision, rather than relying on the category-specific optimization.

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TABLE I

PERFORMANCE COMPARISON ON iSAID-5^i UNDER THE ONE-SHOT AND FIVE-SHOT SETTINGS, UTILIZING MIOU AND FB-IOU (%) AS EVALUATION METRICS. "MEAN" DENOTES THE AVERAGE MIOU ACROSS THE THREEFOLD, AND "FB-IOU" REPRESENTS THE AVERAGE FOREGROUND–BACKGROUND IOU OVER THE SAME FOLDS

Backbone	Method	1-shot					5-shot				
		Fold-0	Fold-1	Fold-2	Mean	FB-IoU	Fold-0	Fold-1	Fold-2	Mean	FB-IoU
ResNet-50	CyCTR (NeurIPS'21) [55]	51.15	38.40	53.79	47.78	62.86	51.91	39.01	54.83	48.58	63.81
	PFENet (TPAMI'22) [80]	51.34	38.79	52.26	47.46	63.34	54.71	41.51	54.45	50.22	64.99
	NERTNet (CVPR'22) [54]	49.52	38.66	51.87	46.68	62.84	50.60	40.99	55.07	48.89	63.74
	DCPNet (IJCV'23) [34]	48.43	37.59	52.09	46.04	62.76	50.34	40.75	51.33	47.47	64.10
	SCCAN (ICCV'23) [57]	52.24	38.60	52.84	47.89	63.26	52.75	38.95	54.26	48.65	62.23
	BAM (TPAMI'23) [81]	58.17	42.12	50.95	50.41	67.03	62.51	43.16	58.24	54.64	67.92
	R2Net (TGRS'23) [76]	56.81	39.85	49.02	48.56	62.65	60.47	41.43	50.24	50.71	65.70
	DMNet (TGRS'23) [27]	54.45	40.68	53.60	49.58	64.46	57.67	41.06	55.28	51.34	65.81
	MGANet (TGRS'24) [78]	55.81	43.52	56.23	51.85	64.49	57.81	45.32	57.87	53.67	65.99
	VRP-SAM (CVPR'24) [82]	52.37	41.62	53.60	49.20	64.55	55.08	43.21	56.95	51.75	65.57
	FCP (AAAI'25) [83]	54.13	41.87	55.96	50.65	66.04	56.93	45.25	59.60	53.93	66.59
	VLP-SAM (arXiv'25) [84]	55.68	43.64	55.74	51.69	66.62	59.37	45.50	59.04	54.64	67.29
	ViRefSAM (Ours)	**57.92**	**44.74**	**56.51**	**53.06**	**67.47**	**61.23**	**46.48**	**59.52**	**55.74**	**68.34**
ResNet-101	CyCTR (NeurIPS'21) [55]	50.89	38.89	52.22	47.33	62.35	52.15	40.28	55.32	49.25	64.45
	PFENet (TPAMI'22) [80]	50.69	38.37	52.85	47.30	62.46	54.40	41.55	50.55	48.83	64.57
	NERTNet (CVPR'22) [54]	50.33	38.73	51.23	46.76	63.25	53.24	41.87	51.53	48.88	64.16
	DCPNet (IJCV'23) [34]	47.63	38.80	49.34	45.26	62.56	50.68	40.02	54.52	48.41	62.96
	SCCAN (ICCV'23) [57]	52.71	38.76	53.45	48.31	62.72	53.85	39.11	54.56	49.17	64.27
	BAM (TPAMI'23) [81]	59.10	41.65	50.54	50.43	66.05	61.57	42.32	55.34	53.08	66.42
	R2Net (TGRS'23) [76]	57.74	39.76	48.45	48.65	63.38	58.83	40.58	49.69	49.70	63.79
	DMNet (TGRS'23) [27]	54.01	40.04	53.57	49.21	64.03	55.70	41.69	56.47	51.29	65.88
	MGANet (TGRS'24) [78]	55.03	42.90	54.69	50.87	64.35	56.43	44.85	57.67	52.98	66.42
	VRP-SAM (CVPR'24) [82]	51.68	41.88	53.52	49.03	65.41	55.36	44.09	56.31	51.92	65.13
	FCP (AAAI'25) [83]	53.55	42.93	55.72	50.73	65.63	57.09	45.46	59.54	54.03	66.81
	VLP-SAM (arXiv'25) [84]	56.86	43.91	56.14	52.30	66.75	60.47	45.82	58.77	55.02	68.20
	ViRefSAM (Ours)	**57.10**	**44.43**	**56.84**	**52.79**	**67.22**	**60.38**	**46.71**	**59.22**	**55.44**	**68.87**
ViT-B/16	FPTrans (NeurIPS'22) [85]	50.04	36.85	53.30	46.73	62.36	57.92	47.42	64.23	56.52	66.12
	AgMTR (IJCV'24) [86]	55.45	40.67	53.81	49.98	63.59	59.05	47.86	64.08	57.00	67.60
	ViRefSAM (Ours)	**56.34**	**42.50**	**55.29**	**51.38**	**65.25**	**59.74**	**48.23**	**63.94**	**57.30**	**68.49**
DeiT-B/16	FPTrans (NeurIPS'22) [85]	51.59	37.79	52.85	47.41	62.52	58.52	47.24	65.37	57.04	67.63
	AgMTR (IJCV'24) [86]	57.51	43.54	53.69	51.58	64.63	61.51	49.23	64.05	58.26	68.72
	ViRefSAM (Ours)	**58.13**	**45.26**	**56.01**	**53.13**	**66.99**	**62.20**	**49.57**	**63.28**	**58.35**	**69.25**

LoveDA-2^i: Table III reports the one-shot and five-shot segmentation results on the LoveDA-2^i dataset. As shown, the overall mIoU scores are relatively lower compared to other benchmarks, primarily due to the inherent domain shift in LoveDA, which includes heterogeneous urban and rural landscapes. Moreover, the dataset focuses on land cover categories rather than well-defined foreground objects (e.g., "plane"), posing greater challenges for model generalization. Despite these difficulties, ViRefSAM consistently achieves superior performance. With the ResNet-50 backbone, ViRefSAM achieves 1.06% and 0.66% higher mIoU than the VLM-based method VLP-SAM [84] under the one-shot and five-shot settings, respectively. Although VLP-SAM leverages an additional text encoder to assist segmentation, ViRefSAM still outperforms it. Moreover, ViRefSAM surpasses the strongest vision-based methods by 1.48% and 1.07% mIoU under the same settings. Similar gains are observed with the ResNet-101 backbone. These results demonstrate the robustness and strong generalization ability of ViRefSAM in handling complex cross-domain RS segmentation scenarios.

COCO-20^i: To further evaluate the generalizability and applicability of the proposed method beyond RS, we conducted experiments on the more commonly used COCO-20^i dataset. Table IV presents the one-shot and five-shot segmentation results. In addition to comparing ViRefSAM with state-of-the-art FSS methods, we also include recent approaches leveraging vision foundation models for FSS, such as SegGPT [95], PerSAM [96], and LLaFS [97]. ViRefSAM achieves a competitive mIoU of 54.13% under the 1-shot setting, outperforming SAM-based methods including PerSAM (23.03%), Matcher (52.73%), and VRP-SAM (53.88%), as well as LLaFS (53.95%), which incorporates large language models. Notably, methods like Painter [98] and SegGPT [95] are trained utilizing all COCO categories, while ViRefSAM achieves strong performance despite being

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TABLE II

FINE-GRAINED CLASSWISE PERFORMANCE COMPARISON ON iSAID-5^i UNDER THE ONE-SHOT SETTING USING IOU AS THE EVALUATION METRIC. C1–C15 DENOTE THE UNSEEN CLASSES ACROSS THE THREE EVALUATION FOLDS, WHERE C1–C5 ARE FROM FOLD 0, C6–C10 FROM FOLD 1, AND C11–C15 FROM FOLD 2. "MEAN" DENOTES THE AVERAGE MIOU ACROSS THE THREEFOLD

Method	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15	Mean
ResNet-50																
ASGNet (CVPR'21) [87]	42.83	50.95	61.00	39.23	48.96	39.75	49.70	33.04	31.96	29.67	72.48	48.73	45.23	24.00	42.83	44.02
CyCTR (NeurIPS'21) [55]	52.20	51.41	60.14	40.01	**51.97**	**45.52**	53.04	35.23	29.89	28.34	**76.77**	49.51	63.12	21.59	**57.98**	47.78
PFENet (TPAMI'22) [80]	50.77	47.05	53.69	63.03	42.13	23.84	48.61	47.96	37.95	35.57	50.98	65.33	55.98	65.79	23.22	47.46
NERTNet (CVPR'22) [54]	49.46	52.34	43.42	61.62	40.76	27.87	45.48	48.87	35.12	35.96	52.78	60.71	57.11	63.95	24.8	46.68
DCPNet (IJCV'23) [34]	47.44	47.54	49.19	63.84	34.13	23.32	47.09	49.49	30.34	**37.72**	44.79	70.2	58.94	64	22.55	46.04
SCCAN (ICCV'23) [57]	**54.03**	52.15	**63.41**	55.14	36.47	32.59	43.50	52.82	30.43	33.64	49.60	70.92	**65.22**	58.50	19.98	47.89
BAM (TPAMI'23) [81]	51.43	64.15	60.98	65.08	49.21	33.64	50.35	56.81	33.47	36.33	50.97	64.79	50.67	**67.48**	20.84	50.41
R2Net (TGRS'23) [34]	45.25	62.54	58.59	**67.54**	50.13	34.30	50.44	49.83	32.11	32.57	36.78	54.32	51.41	62.51	40.08	48.56
DMNet (TGRS'23) [27]	51.41	56.87	56.95	66.03	40.99	33.17	49.18	49.66	36.06	35.33	51.14	**71.12**	50.46	67.01	28.27	49.58
ViRefSAM (Ours)	52.68	**64.32**	60.12	66.87	45.61	37.8	**54.14**	**57.38**	**38.11**	36.27	60.74	69.51	60.48	58.62	33.2	**53.06**
ResNet-101																
ASGNet (CVPR'21) [87]	42.81	50.90	57.23	34.69	52.15	41.00	52.03	35.95	36.22	27.16	66.46	63.57	45.04	21.20	50.15	45.10
CyCTR (NeurIPS'21) [55]	48.76	49.24	58.79	45.06	**52.59**	**52.01**	45.42	33.52	34.82	28.7	**68.74**	43.47	**66.18**	24.37	**58.24**	47.33
PFENet (TPAMI'22) [80]	46.63	48.55	57.02	59.99	41.25	27.32	42.42	48.66	34.63	38.82	54.78	70.40	51.71	64.30	23.05	47.30
NERTNet (CVPR'22) [54]	50.41	48.21	52.63	61.77	38.61	29.17	37.44	53.25	34.79	39.03	41.05	72.40	59.08	61.30	22.31	46.76
DCPNet (IJCV'23) [34]	49.24	39.46	47.46	59.32	42.67	24.16	48.82	46.89	34.74	**39.40**	37.59	64.53	58.92	62.70	22.95	45.26
SCCAN (ICCV'23) [57]	48.70	55.04	57.39	57.34	45.08	30.00	44.29	52.87	31.90	34.73	42.46	**78.50**	61.31	61.54	23.44	48.31
BAM (TPAMI'23) [81]	55.03	**63.48**	62.01	65.57	49.38	25.81	**55.12**	**58.04**	30.82	38.47	58.56	64.01	52.51	**65.60**	12.04	50.43
R2Net (TGRS'23) [34]	**56.33**	61.36	**68.92**	60.11	41.97	34.86	45.39	51.31	28.47	38.74	52.25	67.48	48.73	60.36	13.43	48.65
DMNet (TGRS'23) [27]	51.98	56.31	56.98	**67.13**	37.65	32.97	48.21	49.23	34.25	35.54	45.89	70.04	57.86	65.50	28.56	49.21
ViRefSAM (Ours)	53.65	62.87	61.21	64.34	43.43	36.77	54.95	56.24	**38.47**	35.72	62.28	68.54	61.33	59.07	32.98	**52.79**
DeiT-B/16																
FPTrans (NeurIPS'22) [85]	44.56	63.97	54.83	67.54	27.07	27.49	49.24	45.93	37.66	28.62	45.40	**84.63**	41.39	66.75	26.11	47.41
AgMTR (IJCV'24) [86]	49.22	**68.55**	58.99	**70.37**	40.41	30.73	**61.07**	50.55	37.37	**37.99**	30.08	83.14	56.73	**67.26**	**31.23**	51.58
ViRefSAM (Ours)	**52.35**	67.49	**60.21**	67.79	**42.81**	**34.40**	57.49	**58.21**	**40.53**	35.67	**48.73**	79.55	**57.48**	64.62	29.67	**53.13**

Note that the categories of C1-C15 are: "ship", "storage tank", "baseball diamond", "tennis court", "basketball court", "ground track field", "bridge", "large vehicle", "small vehicle", "helicopter", "swimming pool", "roundabout", "soccer ball field", "plane", "harbor".

TABLE III

PERFORMANCE COMPARISON ON LOVEDA-2^i UNDER THE ONE-SHOT AND FIVE-SHOT SETTINGS, UTILIZING MIOU AND FB-IOU (%) AS EVALUATION METRICS. "MEAN" DENOTES THE AVERAGE MIOU ACROSS THE THREEFOLD, AND "FB-IOU" REPRESENTS THE AVERAGE FOREGROUND–BACKGROUND IOU OVER THE SAME FOLDS

Backbone	Method	1-shot					5-shot				
		Fold-0	Fold-1	Fold-2	Mean	FB-IoU	Fold-0	Fold-1	Fold-2	Mean	FB-IoU
ResNet-50	SCL (CVPR'21) [91]	15.14	20.45	25.00	20.20	24.60	14.25	21.09	23.65	19.66	24.69
	ASGNet (CVPR'21) [87]	15.91	20.21	22.33	19.48	36.39	18.38	26.29	36.34	27.00	39.59
	CyCTR (NeurIPS'21) [55]	13.17	23.43	21.99	19.53	38.47	13.81	27.40	26.15	22.45	42.71
	PFENet (TPAMI'22) [80]	17.13	22.20	26.49	21.94	33.48	15.83	25.73	24.74	22.10	34.77
	NERTNet (CVPR'22) [54]	16.05	22.69	21.87	20.20	32.67	15.79	24.94	23.42	21.38	31.79
	DCPNet (IJCV'23) [34]	16.67	23.10	24.44	21.40	36.36	13.43	25.59	28.06	22.36	33.89
	DMNet (TGRS'23) [27]	19.29	25.52	31.53	25.45	43.40	24.62	33.80	33.12	30.51	50.93
	MGANet (TGRS'24) [78]	20.55	27.51	33.30	27.12	44.91	26.53	35.99	36.69	33.07	50.68
	FCP (AAAI'25) [83]	20.32	26.85	33.68	26.95	44.26	25.90	34.47	37.25	32.54	50.17
	VLP-SAM (arXiv'25) [84]	21.54	27.60	33.49	27.54	44.83	27.75	35.58	37.11	33.48	50.33
	ViRefSAM (Ours)	**22.09**	**28.94**	**34.77**	**28.60**	**45.51**	**28.68**	**36.25**	**37.48**	**34.14**	**51.80**
ResNet-101	SCL (CVPR'21) [91]	15.62	17.87	25.54	19.68	31.03	14.96	20.26	24.02	19.75	31.57
	ASGNet (CVPR'21) [87]	14.65	19.90	25.43	19.99	30.89	18.00	25.43	36.00	26.48	39.26
	CyCTR (NeurIPS'21) [55]	13.16	20.63	20.55	18.11	38.94	15.40	25.27	22.01	20.89	36.03
	PFENet (TPAMI'22) [80]	15.83	25.73	24.74	22.10	35.65	15.62	24.37	26.64	22.21	36.08
	NERTNet (CVPR'22) [54]	15.51	19.65	30.07	21.74	33.68	15.18	23.50	31.83	23.50	32.16
	DCPNet (IJCV'23) [34]	16.52	20.20	33.61	23.44	36.57	16.97	25.08	25.10	22.38	35.71
	DMNet (TGRS'23) [27]	21.18	24.57	28.74	24.83	47.47	24.65	33.97	36.89	31.84	53.38
	MGANet (TGRS'24) [78]	22.97	27.79	28.14	26.30	46.83	26.29	34.88	36.84	32.67	52.88
	FCP (AAAI'25) [83]	22.75	27.12	28.54	26.14	44.18	25.67	33.35	37.41	32.14	50.37
	VLP-SAM (arXiv'25) [84]	23.26	27.35	31.57	27.39	44.79	27.30	34.93	36.04	32.76	51.82
	ViRefSAM (Ours)	**23.53**	**29.82**	**31.62**	**28.32**	**46.12**	**27.45**	**35.74**	**36.66**	**33.28**	**53.45**

evaluated on entirely unseen categories, highlighting its robust generalization capabilities under the few-shot setting.

2) Qualitative Results: To further demonstrate the superiority of the proposed ViRefSAM, we visualize representative segmentation results across three FSS datasets. As depicted in Fig. 7, ViRefSAM achieves high-quality segmentation on iSAID-5^i, clearly outperforming existing methods. Compared to current FSS methods such as R2Net [76], our method more

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TABLE IV

PERFORMANCE COMPARISON ON COCO-20^i UNDER THE ONE-SHOT AND FIVE-SHOT SETTINGS, UTILIZING MIOU (%) AS EVALUATION METRICS. "MEAN" DENOTES THE AVERAGE MIOU ACROSS THE FOUR FOLDS. BOLD AND UNDERLINED VALUES THE INDICATE THE BEST AND SECOND-BEST PERFORMANCE, RESPECTIVELY. NOTABLY, PAINTER [98] AND SEGGPT [95] WERE TRAINED WITH ACCESS TO ALL COCO CATEGORIES, WHILE THE OTHER METHODS MAINTAIN EVALUATIONS ON UNSEEN CLASSES

Method		Fold-0	Fold-1	Fold-2	Fold-3	Mean
Specialized FSS Model	DBMNet (TIP'24) [105]	41.80	45.60	43.20	41.30	42.98
	HMNet (NeurIPS'24) [106]	45.50	58.70	52.90	51.40	52.13
	AgMTR (IJCV'24) [86]	44.01	55.51	49.90	46.61	49.01
Foundation Model	Painter (CVPR'23) [98]	31.20	35.30	33.50	32.40	33.10
	SegGPT (ICCV'23) [95]	56.30	57.40	58.90	51.70	56.08
	PerSAM (ICLR'24) [96]	23.10	23.60	22.00	23.40	23.03
	PerSAM-F (ICLR'24) [96]	22.30	24.00	23.40	24.10	23.45
	Matcher (ICLR'24) [107]	**52.70**	53.50	52.60	52.10	52.73
	VRP-SAM (CVPR'24) [82]	48.10	55.80	**60.00**	51.60	53.88
	PAT (TPAMI'24) [108]	40.62	51.94	48.99	50.66	48.05
	LLaFS (CVPR'24) [97]	47.50	**58.80**	56.20	**53.30**	53.95
	FCP (AAAI'25) [83]	46.40	56.40	55.30	51.80	52.50
	ViRefSAM (Ours)	49.41	56.17	59.19	51.74	**54.13**

accurately localizes and delineates the target objects while suppressing irrelevant regions. For instance, in the second column, ViRefSAM effectively segments densely arranged "ships," while R2Net fails to capture precise object boundaries. Similarly, in the third column, our method accurately segments the roundabout structure, while R2Net incorrectly activates irrelevant categories such as "parking lots." These improvements are primarily attributed to the proposed VCP encoder, which effectively leverages reference images to activate SAM's prompt-driven generalization capabilities. Moreover, the DTA adapter further enhances SAM's focus on task-relevant semantics, resulting in more precise segmentation. Figs. 8 and 9 show qualitative results on LoveDA-2^i and COCO-20^i, respectively. ViRefSAM consistently delivers robust performance across diverse and challenging domains, confirming its strong generalization ability and applicability to both cross-domain RS tasks and natural image scenarios.

3) Comparison With VLM-Based Methods: To align with the emergence of vision language models (VLMs) in the AI 2.0 era, we compared ViRefSAM with recent VLM-based FSS methods on the iSAID-5^i dataset, including PGMA-Net [70], PI-CLIP [71], MVLPNet [72], and VLP-SAM [84]. As depicted in Table V, ViRefSAM outperforms all comparable methods under the one-shot setting, achieving the highest mIoU (53.13%) and FB-IoU (67.47%) with the DeiT-B/16. Under the five-shot setting, ViRefSAM still maintains strong performance, surpassing other methods in mIoU and achieving a good balance between both metrics. These results highlight the effectiveness of ViRefSAM in RS segmentation, demonstrating its generalization ability even as an unimodal model.

4) Performance on Dense Small Objects: To further evaluate the model's performance in dense small-object scenarios, we conducted a category-specific comparison on "ship," "small vehicle," and "large vehicle" classes from the iSAID-5^i dataset. As shown in Table VI, ViRefSAM achieves the highest scores across all categories in both IoU and boundary IoU (BIoU) [99]. Notably, the boundary-level improvement in BIoU indicates that ViRefSAM generates more precise object contours, particularly in densely distributed regions with strong interobject interference. This improvement is not merely attributed to the SAM backbone itself (as the SAM+PFENet baseline performs suboptimally);

TABLE V

PERFORMANCE COMPARISON BETWEEN VIREFSAM AND RECENT VLM-BASED FSS METHODS ON ISAID-5^i UNDER THE ONE-SHOT AND FIVE-SHOT SETTINGS, UTILIZING MIOU AND FB-IOU (%) AS EVALUATION METRICS. THE BEST PERFORMANCE IN EACH COLUMN IS MARKED IN BOLD, WITH THE SECOND-BEST PERFORMANCE UNDERLINED

Method	Backbone	1-shot		5-shot	
		mIoU	FB-IoU	mIoU	FB-IoU
PGMA-Net [70]	CLIP+ResNet-50	47.04	59.69	50.74	63.17
	CLIP+ResNet-101	49.71	61.85	53.68	67.06
PI-CLIP [71]	CLIP+VGG-16	45.97	60.23	50.08	63.26
	CLIP+ResNet-50	49.23	62.03	51.82	65.34
MVLPNet [72]	CLIP+VGG-16	48.87	64.69	54.52	66.75
	CLIP+ResNet-50	51.80	67.34	56.09	**70.45**
VLP-SAM [84]	CLIP+ResNet-50	51.69	66.62	54.64	67.29
	CLIP+ResNet-101	52.30	66.75	55.02	68.20
ViRefSAM (Ours)	ResNet-50	53.06	**67.47**	55.74	68.34
	ViT-B/16	51.38	65.25	57.30	68.49
	DeiT-B/16	**53.13**	66.99	**58.35**	69.25

TABLE VI

PERFORMANCE COMPARISON OF VIREFSAM AND OTHER SAM-BASED SEGMENTATION METHODS ON DENSE SMALL-OBJECT CATEGORIES ("SHIP," "SMALL VEHICLE," AND "LARGE VEHICLE") FROM THE ISAID-5^i DATASET. BOTH IOU AND BIOU [99] ARE REPORTED TO JOINTLY EVALUATE OVERALL OVERLAP AND BOUNDARY PRECISION. "OVERALL" DENOTES THE AVERAGE PERFORMANCE ACROSS ALL CATEGORIES

Method	C1:ship		C8:small vehicle		C9:large vehicle		Overall	
	IoU	BIoU	IoU	BIoU	IoU	BIoU	IoU	BIoU
AgMTR [86]	49.22	46.23	50.55	47.11	37.37	32.64	51.58	46.90
SAM+PFENet [80]	51.79	47.12	48.97	44.75	37.96	33.37	48.47	42.84
VRP-SAM [82]	46.80	43.64	54.26	50.73	35.62	29.94	49.20	44.57
ViRefSAM	**52.68**	**48.25**	**57.38**	**53.55**	**38.11**	**34.93**	**53.06**	**49.26**

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TABLE VII

EXTENDED EXPERIMENTS OF CROSS-DOMAIN FSS. NOTABLY, THE PROPOSED MODEL IS TRAINED ON THE CV SEGMENTATION DATASET PASCAL VOC [100] AND EVALUATED ON THE RS SEGMENTATION DATASET DEEPGLOBE [101]. SOME COMPARISON RESULTS ARE FROM [102]

Method	1-shot	5-shot
HDMNet (CVPR'23) [58]	25.40	39.10
RestNet (BMVC'23) [103]	22.70	29.90
PMNet(WACV'24) [104]	37.10	41.60
ABCDFSS (CVPR'24) [102]	42.60	49.00
AgMTR (IJCV'24) [86]	41.42	47.11
ViRefSAM (Ours)	**43.75**	**49.98**

rather, we attribute the gains to the integration of the VCP Encoder and the DTA adapter. The former effectively captures fine-grained contextual relationships between reference and target images, while the latter dynamically injects object-aware clues into SAM's visual representations, thereby achieving more accurate localization and reducing false activations in complex RS scenes.

5) Extension in Cross-Domain FSS: To further investigate the generalization capability of the proposed ViRefSAM across unseen domains in real-world applications, we conduct a cross-domain FSS experiment, in which the samples utilized for training and testing are from distinct domains. Specifically, the model is trained on the general-domain PASCAL VOC dataset [100] and evaluated on the RS dataset DeepGlobe [101], following the cross-domain protocol described in [102]. As presented in Table VII, ViRefSAM achieves 43.75% and 49.98% mIoU under the one-shot and five-shot settings, respectively, outperforming all compared methods. These results highlight ViRefSAM's strong cross-domain transferability and its ability to bridge the substantial visual and semantic gap between natural and aerial imagery, demonstrating its potential as a robust and adaptable framework for RS segmentation under limited supervision.

C. Ablation Study and Analysis

We conduct a series of ablation studies to further investigate and understand the effectiveness of the proposed method. Unless otherwise specified, all experiments in this section are performed on the iSAID-5^i dataset under the one-shot setting using the ResNet-50 backbone.

1) Performance Under Various Reference Annotation Schemes: To further alleviate the burden of dense annotations in real-world scenarios, this section explores the impact of different annotation schemes for reference images on segmentation performance, including more economical alternatives such as points, bounding boxes, and scribbles. Following the setup in [32] and [82], these annotations are randomly generated from the original dense masks of the reference images. Fig. 10 and Table VIII present the quantitative and qualitative results under the one-shot setting on three FSS benchmark datasets.

It can be observed that ViRefSAM achieves mIoU scores of 50.11%, 21.98%, and 49.32% on the three datasets using scribble annotations, which are even comparable to several FSS methods relying on dense masks, such as PFENet [80]. This demonstrates the strong robustness of ViRefSAM to annotation quality. In contrast, bounding box annotations yield noticeably lower performance due to the background noise they introduce, which hinders the extraction of discriminative semantics from the reference image. Scribble annotations, though containing fewer foreground pixels, provide more accurate spatial clues and effectively suppress irrelevant background regions, enabling the model to focus on class-relevant features. As shown in Fig. 10, this advantage is most pronounced for irregularly shaped targets (e.g., "planes"), where bounding boxes include excessive nonobject pixels, whereas scribble annotations produce cleaner and sharper boundaries. Overall, scribble annotations strike a favorable balance between annotation cost and segmentation accuracy, validating their practicality in real-world RS applications.

2) Component Analysis: This section investigates the performance contributions of different components in ViRefSAM through ablation studies. Specifically, we compare three variants: reused prompt-SAM (RP-SAM), the PFENet-based baseline, and our ViRefSAM.

We first introduce RP-SAM, a straightforward prompt reuse scheme commonly adopted in practical settings. As illustrated in Fig. 1(b), RP-SAM directly reuses the prompts (e.g., points or bounding boxes) annotated on the reference image. These prompts are encoded by SAM's prompt encoder and then applied to the target image through the SAM mask decoder to generate the segmentation result.

Next, we define the baseline, which integrates SAM with the existing FSS method PFENet [80]. As illustrated in Fig. 11, given the target image and the reference image, both are input into PFENet to generate a pseudo mask for the target image. Various prompt types (e.g., point, box, and mask) are then randomly sampled from this predicted mask and encoded by SAM's prompt encoder. The resulting prompt embeddings are passed to the SAM mask decoder to produce the final segmentation result.

Table IX summarizes the quantitative results of different ViRefSAM variants under the one-shot setting on iSAID-5^i. Overall, we observe the following.

1) Compared with reused prompt-SAM, which directly transfers prompts from the reference image to target images without adaptation, our ViRefSAM achieves significantly better performance. Specifically, the full version of ViRefSAM improves mIoU by 7.68% over Reused Prompt-SAM, indicating that static prompt reuse is insufficient to handle the large semantic variations in RS scenes, and emphasizing the necessity of adaptive prompt generation guided by reference semantics.
2) By incorporating the VCP Encoder, our model achieves a 3.14% mIoU improvement over the baseline. This clearly demonstrates the effectiveness of the VCP encoder and its better compatibility with the SAM architecture.

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TABLE VIII

ABLATION STUDY ON THE IMPACT OF DIFFERENT REFERENCE ANNOTATION SCHEMES ACROSS THREE FSS DATASETS UNDER THE ONE-SHOT SETTING. THE MIOU (%) IS UTILIZED AS THE EVALUATION METRIC. BOLD AND UNDERLINED VALUES INDICATE THE BEST AND SECOND-BEST PERFORMANCE, RESPECTIVELY

Annotation Scheme	iSAID-5^i				LoveDA-2^i				COCO-20^i				
	Fold-0	Fold-1	Fold-2	Mean	Fold-0	Fold-1	Fold-2	Mean	Fold-0	Fold-1	Fold-2	Fold-3	Mean
Point	45.77	38.69	48.26	44.24	11.58	16.59	17.37	15.18	32.40	42.63	44.18	39.98	39.80
Scribble	<u>56.25</u>	<u>41.75</u>	<u>52.33</u>	<u>50.11</u>	<u>18.29</u>	22.19	<u>25.45</u>	<u>21.98</u>	44.75	<u>51.58</u>	53.63	47.30	49.32
Box	55.30	40.18	51.42	48.97	17.43	<u>22.22</u>	24.91	21.52	<u>45.21</u>	50.62	<u>54.49</u>	<u>48.13</u>	<u>49.61</u>
Mask	**57.92**	**44.74**	**56.51**	**53.06**	**22.09**	**28.94**	**34.77**	**28.60**	**49.41**	**56.17**	**59.19**	**51.74**	**54.13**

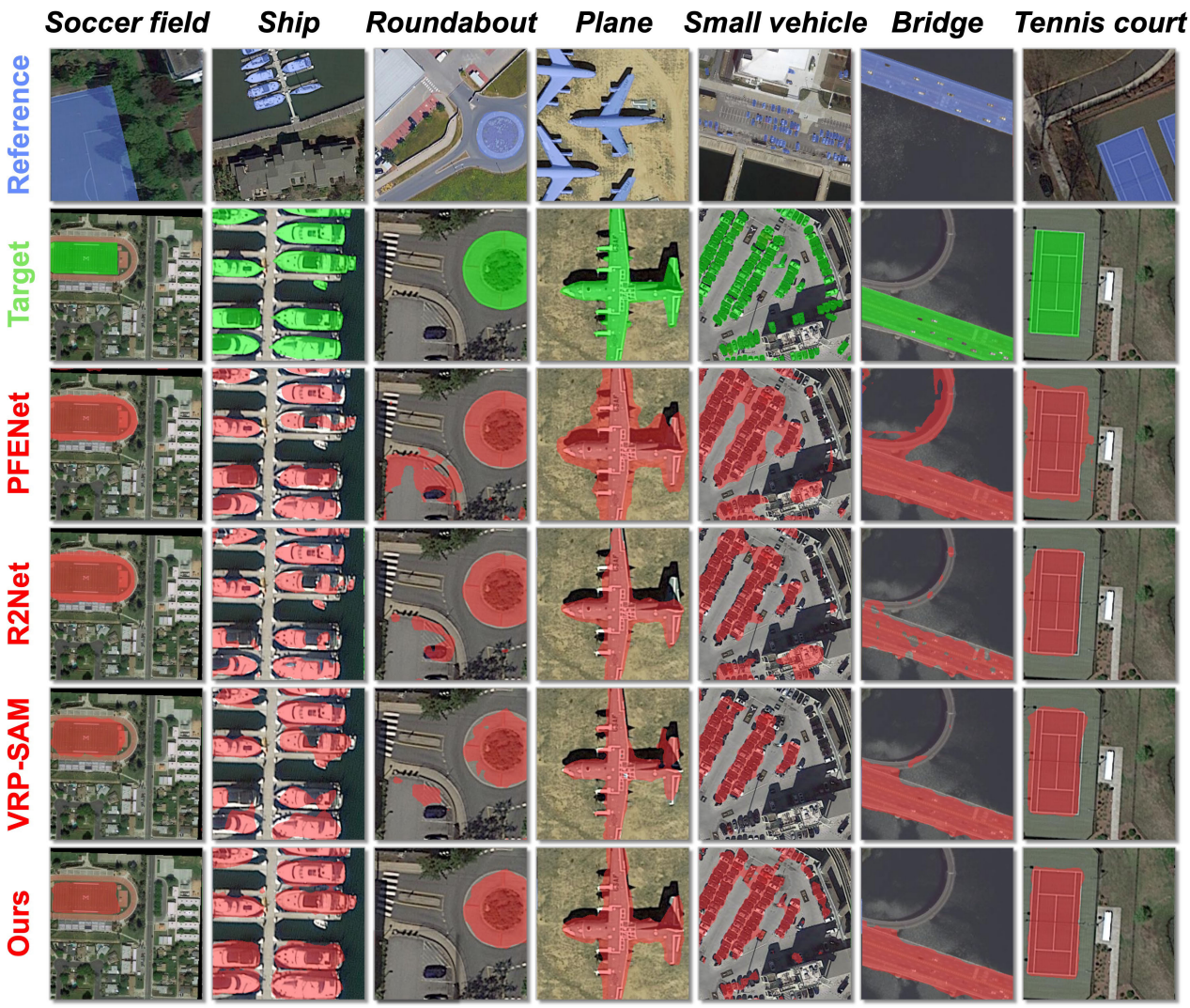


Fig. 7. Qualitative visualization of our ViRefSAM and several competing methods under the one-shot setting on iSAID-5^i benchmark.(Top to bottom) Each row shows: 1) reference images with GT masks; 2) target images with ground-truth (GT) masks; 3) segmentation results of R2Net; 4) segmentation results of VRP-SAM; and 5) segmentation results of our ViRefSAM.

3) Introducing the DTA Adapter further improves performance by an additional 1.45% mIoU, highlighting its ability to inject task-specific semantic clues into SAM's image encoder and guide the model to focus on target-specific regions in RS images while maintaining generalization.

4) As presented in Fig. 12, we also visualize segmentation results under different variants. With progressive integration of each component, the results show increasingly precise segmentation with better object focus and fewer false activations, qualitatively validating the effectiveness and necessity of the proposed components.

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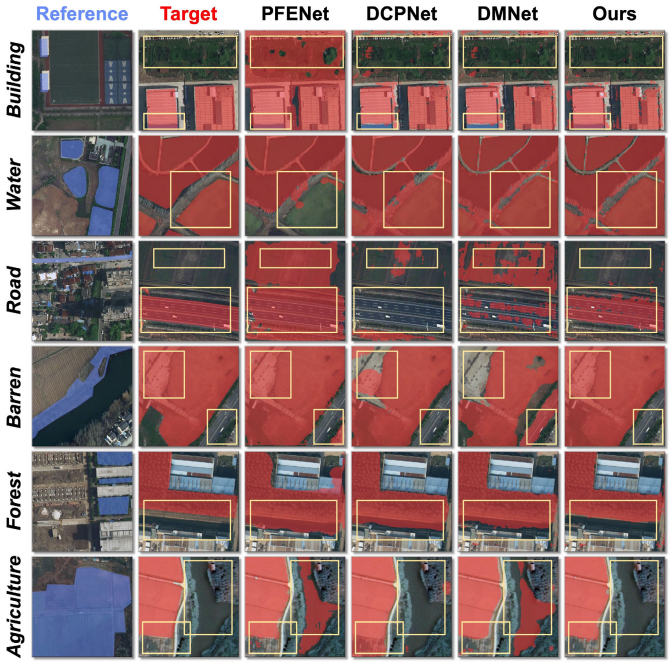


Fig. 8. Qualitative visualization of our ViRefSAM and several competing methods under the one-shot setting on LoveDA-2^i benchmark. (Left to right) Each column shows: 1) reference images with GT masks; 2) target images with GT masks; 3) segmentation results of PFENet; 4) segmentation results of DCPNet; 5) segmentation results of DMNet; and 6) segmentation results of our ViRefSAM.



Fig. 9. Qualitative visualization of our ViRefSAM under the one-shot setting on COCO-20^i benchmark. (Top to bottom) Each row shows: 1) reference images with GT masks; 2) target images with GT masks; and 3) segmentation results of our ViRefSAM.

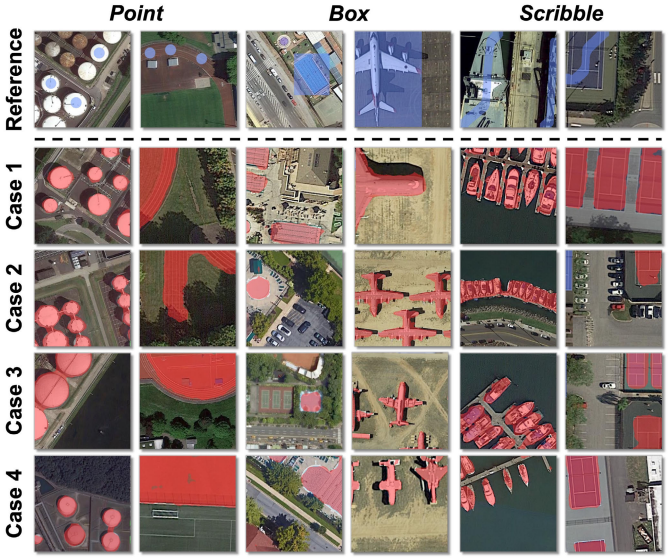


Fig. 10. Qualitative visualization of different reference annotation schemes on the iSAID-5^i benchmark under the one-shot setting. (Top to bottom) Reference images with GT annotations and four segmentation cases are provided.

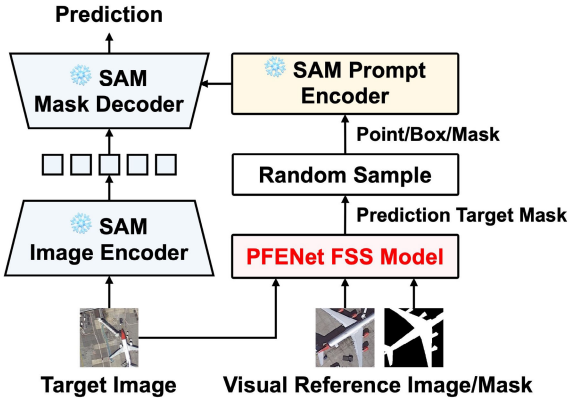


Fig. 11. Baseline architecture of proposed ViRefSAM. It combines SAM with PFENet, where PFENet generates pseudo-masks from reference images. Various prompt types (e.g., points, boxes, and masks) are randomly sampled from these predicted masks and passed through SAM's prompt encoder to generate prompt embeddings, which are then utilized by the SAM mask decoder to produce the final segmentation of the target image.

3) Ablation Study of the VCP Encoder: We first investigate the effect of different variants of the visual contextual encoder on segmentation performance. As presented in Table X, by comparing case (c) with case (a), and case (c) with case (b), we observe that both the visual context interaction and the object-aware prompt generation effectively enhance the segmentation performance of SAM, yielding mIoU improvements of 1.12% and 0.63%, respectively. These results validate the effectiveness of the proposed components: the former captures class-specific clues from both reference and target images, while the latter generates object-aware embeddings aligned with the mask decoder.

Next, we analyze the sensitivity of two key hyperparameters: the number of learnable queries (N_q) and the number of background prototypes (N_b). As shown in Fig. 13 and Table XI, increasing N_q from 1 to 64 steadily improves segmentation performance, with the highest mIoU (53.06%) achieved at $N_q = 64$, beyond which the gain saturates. Similarly, the model performs best when $N_b = 5$, while larger values lead to redundant prototypes and slight performance degradation. Hence, we set $N_q = 64$ and $N_b = 5$ in all experiments to achieve a balance between accuracy and efficiency.

4) Ablation Study of the DTA Adapter: In this section, we discuss the impact of different fine-tuning strategies applied to the SAM image encoder on segmentation performance, including parameter freezing, full-parameter tuning, standard adapter tuning [31], LoRA tuning [109], ConvPass tuning [110], and the proposed DTA adapter tuning. The results are summarized in Table XII. The key observations are as follows.

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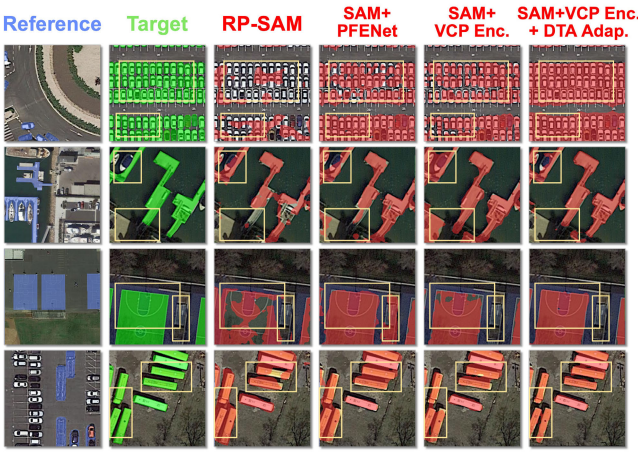


Fig. 12. Qualitative visualization of ViRefSAM variants. (Left to right) Each column shows reference images with GT, target image with GT, segmentation results from reused prompt-SAM (RP-SAM), SAM+PFENet, SAM+VCP Encoder, and SAM+VCP Encoder+DTA Adapter (i.e., the full ViRefSAM).

TABLE IX

ABLATION STUDY OF DIFFERENT COMPONENTS IN VIREFSAM UNDER THE ONE-SHOT SETTING. "VCP ENC." REFERS TO THE VISUAL CONTEXTUAL PROMPT ENCODER, AND "DTA ADAP." DENOTES THE DTA ADAPTER. "SAMPLE PROMPTS" INDICATES THE TYPES OF PROMPTS (E.G., POINTS, BOXES, OR THEIR COMBINATION) THAT ARE FED INTO THE SAM'S MASK DECODER FOR SEGMENTATION

#	Methods	Sample Prompts	mIoU(%)
(a)	Reused Prompt-SAM (Fig.1(b))	Point	45.24
		Box	43.80
		Point+Box	45.38
(b)	SAM+PFENet (Baseline in Fig.11)	Point	48.12
		Box	47.95
		Point+Box	48.47
(c)	SAM+VCP Enc.	-	51.61
(d)	SAM+PFENet+DTA Adap.†	Point+Box	51.32
(e)	SAM+VCP Enc. +DTA Adap.	-	**53.06**

Note †: In this variant, PFENet is integrated in place of the proposed VCP Encoder. Accordingly, the DTA adapter is reconfigured to fuse target features with reference prototypes and PFENet's predicted prior mask, ensuring compatibility within the modified framework.

1) Fully fine-tuning the SAM image encoder leads to a significant performance degradation of 4.42% mIoU. This may be attributed to the overfitting of the encoder to the known classes in the training set, thereby limiting its generalization ability to unseen classes.
2) The standard adapter and LoRA yield no clear improvement, whereas ConvPass provides only a marginal gain, indicating that simply introducing learnable parameters or low-rank projections into the SAM encoder is insufficient for capturing transferable RS knowledge for unseen object segmentation.
3) The proposed DTA adapter improves segmentation performance by 1.45% mIoU. It injects class-specific semantic information from the current task into the SAM image features, enabling the model to dynamically focus on task-relevant objects. This facilitates effective knowledge adaptation in RS scenarios while preserving generalization ability. The qualitative results in Fig. 14 further support this finding.

TABLE X

ABLATION STUDY ON DIFFERENT VARIANTS OF THE PROPOSED VISUAL CONTEXTUAL ENCODER UNDER THE ONE-SHOT SETTING. BOTH VISUAL CONTEXTUAL INTERACTION AND OBJECT-AWARE PROMPT GENERATION IMPROVE SAM'S SEGMENTATION PERFORMANCE

#	Visual Contextual Prompt Encoder		mIoU(%)
	Visual Contextual Interaction	Object-aware Prompt Generation	
(a)	✗	✓	51.94
(b)	✓	✗	52.43
(c)	✓	✓	**53.06**

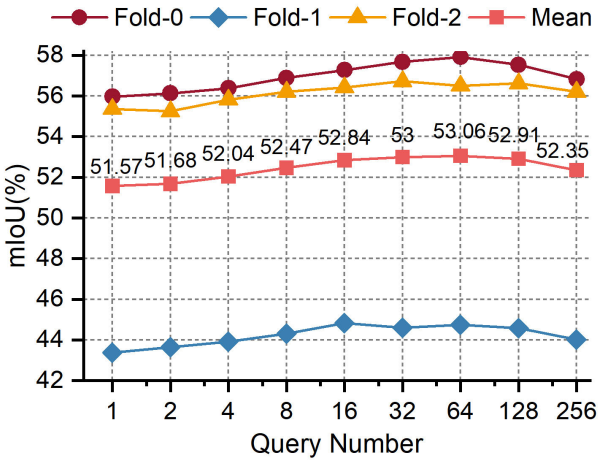


Fig. 13. Ablation study on the number of learnable queries (i.e., N_q) in the object-aware prompt generation from VCP encoder. The best segmentation performance is achieved at $N_q = 64$.

TABLE XI

ABLATION STUDY ON THE NUMBER OF BACKGROUND PROTOTYPES (N_b) IN THE VISUAL CONTEXTUAL INTERACTION FROM VCP ENCODER

N_b	1	3	5	7	9
mIoU	52.37	52.78	**53.06**	52.84	52.65

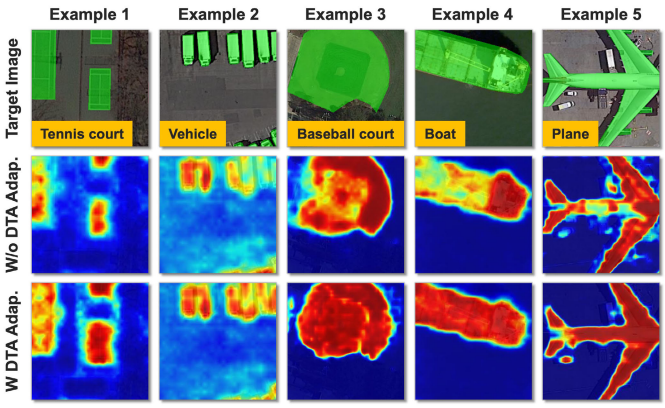


Fig. 14. Qualitative visualization of the DTA adapter. Five segmentation examples illustrate that the DTA adapter effectively guides SAM to focus on class-specific semantic regions in the current task, leading to more accurate object delineation.

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TABLE XII

ABLATION STUDY ON DIFFERENT FINE-TUNING STRATEGIES FOR THE SAM IMAGE ENCODER, INCLUDING PARAMETER FREEZING, FULL-PARAMETER TUNING, ORIGINAL ADAPTER TUNING, AND THE PROPOSED DTA ADAPTER TUNING. Δ INDICATES THE MIOU IMPROVEMENT OVER THE PARAMETER FREEZING BASELINE

#	Fine-tuning Strategy	mIoU(%)	Δ
(a)	Freezing	51.61	-
(b)	Full-parameter Tuning	47.19	-4.42
(c)	Adapter Tuning [31]	51.25	-0.36
(d)	LoRA Tuning [109]	51.50	-0.11
(e)	ConvPass Tuning [110]	51.94	0.33
(f)	DTA Adapter w/o Position Encoding	52.23	0.62
(g)	DTA Adapter Tuning	**53.06**	1.45

TABLE XIII

ABLATION STUDY OF LOSS FUNCTION IN VIREFSAM UNDER THE ONE-SHOT SETTING. THE RESULTS DEMONSTRATE THE CONTRIBUTIONS OF BCE LOSS, DICE LOSS, AND THE REGULARIZATION LOSS IN IMPROVING SEGMENTATION PERFORMANCE AND PROMPT DIVERSITY

#	BCE Loss	Dice Loss	Regularization	mIoU(%)
(a)	✓	✗	✗	52.18
(b)	✗	✓	✗	52.25
(c)	✓	✓	✗	52.61
(d)	✓	✓	✓	**53.06**

TABLE XIV

ABLATION STUDY ON THE WEIGHT OF REGULARIZATION LOSS, I.E., γ IN (10)

γ	0.0	0.1	0.2	0.4	0.6	0.8	1
mIoU	52.61	52.74	**53.06**	52.69	52.50	52.25	51.89

4) In addition, introducing position encoding into the DTA adapter promotes better alignment between reference prototypes and target features, which further helps the model distinguish foreground and background regions effectively.

5) Ablation Study of Loss Function: This section analyzes the impact of loss function design in ViRefSAM on segmentation performance. Table XIII reports the experimental results on the iSAID-5^i dataset under the one-shot setting. The key observations are summarized as follows.

1) A comparison between case (a) and case (c) shows that both the BCE loss and Dice loss contribute positively to segmentation performance. Specifically, the BCE loss facilitates pixel-wise classification, while the Dice loss emphasizes region-level consistency. When combined, they provide complementary advantages and lead to further performance improvements.
2) Introducing a regularization loss (5) in the object-aware prompt generation encourages the generated embeddings to be more semantically diverse and discriminative, effectively reducing semantic redundancy. Furthermore, as shown in Table XIV, setting the regularization weight to $\gamma = 0.2$ in (10) achieves the best tradeoff between regularization strength and segmentation accuracy, whereas larger values slightly degrade performance due to over-regularization.

TABLE XV

ABLATION STUDY OF DIFFERENT SAM ARCHITECTURE VARIANTS IN VIREFSAM, SHOWING VIT-HUGE ACHIEVES THE HIGHEST SEGMENTATION ACCURACY WHILE VIT-BASE OFFERS THE FASTEST INFERENCE SPEED

Backbone	Encoder	mIoU	FB-IoU	Inference seed (FPS)
SAM	ViT-Base	49.42	63.79	29.51
	ViT-Large	51.89	66.50	22.68
	ViT-Huge	53.06	67.47	17.32

TABLE XVI

ABLATION STUDY OF DIFFERENT SAM BACKBONE VARIANTS IN VIREFSAM, SHOWING THAT INTEGRATING SAM-SPT [111] AND ROS-SAM [61] AS THE BACKBONE FURTHER IMPROVES SEGMENTATION ACCURACY COMPARED TO THE ORIGINAL SAM, DEMONSTRATING VIREFSAM'S ADAPTABILITY TO ENHANCE SAM ARCHITECTURES

Method	Backbone	mIoU	FB-IoU
ViRefSAM	Standard SAM [17]	53.06	67.47
	SAM-SPT (AAAI'25) [111]	54.49	68.82
	ROS-SAM (CVPR'25) [61]	**55.28**	**69.65**

6) Ablation Study of SAM Structure Variants: Table XV presents the impact of different SAM architectures on segmentation performance. It can be observed that using the ViT-Huge encoder achieves the highest segmentation accuracy, followed by ViT-Large, indicating that models with larger parameter counts tend to have stronger generalization ability and feature representation capacity. In addition, the ViT-Base model offers the fastest inference speed, demonstrating a favorable trade-off between accuracy and efficiency. Beyond the encoder scale, we further investigate the influence of different SAM backbone variants integrated into ViRefSAM. As shown in Table XVI, replacing the original SAM [17] with SAM-SPT [111] and ROS-SAM [61] yields consistent improvements in both mIoU and FB-IoU. This demonstrates that ViRefSAM can flexibly adapt to enhanced SAM variants while maintaining compatibility with different architectural scales, further validating its generality and extensibility.

D. Failure Case Analysis

Despite the strong performance of ViRefSAM, certain challenging scenarios still lead to segmentation failures, as shown in Fig. 15. These issues primarily arise from ambiguities in object definition, such as in case (a), where a "road" is misclassified as a "bridge" (both constructed from the same material), and in case (b), where resolution differences between the reference and target images cause segmentation failure. In addition, in case (c), object overlap, particularly when "small vehicles" are placed on larger structures, and in case (d), partial occlusion caused by shadows, further complicate accurate segmentation. These examples highlight the model's limitations in handling complex visual environments. In the future, integrating the reasoning capabilities of

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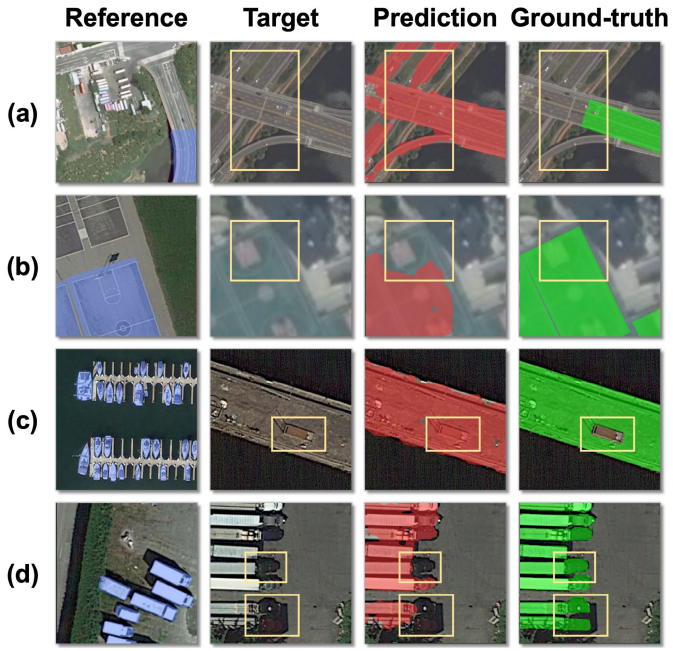


Fig. 15. Failure cases of the proposed ViRefSAM. (Top to bottom) (a) Ill-defined "bridge" that is actually a "road on land," (b) segmentation failure caused by resolution differences between the reference and target images, (c) overlapping "small vehicles" on the "ship," and (d) "large vehicle head" partially obscured by shadow.

large language models with SAM's fine-grained segmentation ability could potentially address some of these challenges, particularly those related to contextual understanding, target overlap, and ambiguous object definitions.

VI. Conclusion

This work presents ViRefSAM, a novel visual reference-guided framework that enhances the SAM for RS segmentation. By leveraging several annotated reference images instead of manually crafted prompts, ViRefSAM enables automatic and class-consistent object segmentation across complex RS scenes without altering SAM's original architecture. The framework integrates two complementary components: the VCP encoder, which provides class-specific prompts at the decoding stage, and the DTA adapter, which injects category-relevant clues into the encoding stage. Together, they guide SAM to focus on task-relevant objects, thereby improving segmentation efficiency and domain adaptability. Extensive experiments on three FSS benchmarks, iSAID-5^i, LoveDA-2^i, and COCO-20^i, demonstrate that ViRefSAM consistently outperforms existing FSS methods in both accuracy and generalization. Beyond its strong empirical results, this study provides a new perspective on integrating large vision foundation models with RS domain knowledge, bridging the gap between general pretraining and RS-specific semantic understanding. Future work will explore multitask learning to further enhance ViRefSAM's scalability and applicability in real-world Earth observation and intelligent interpretation scenarios.

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